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MSc in Computer Science

Explainable AI for brain-wide weighted inference in Electroencephalography

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Explainable AI for brain-wide weighted inference in Electroencephalography
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Abstract

This thesis investigates a graph-based autoencoder (GAE) as a data-driven weighting method for electroencephalography (EEG) analysis. Instead of assuming equal data quality across channels and subjects, the GAE learns subject-level and channel-level weights by reconstructing general linear model (GLM) parameters of event-related potential (ERP) over adjacent structures of the brain. Integrating these learned weights into the hierarchical linear model (LIMO EEG) improves the statistical stability of multiple ERPs, as demonstrated by our investigation of 4 classic components: N170, P3, MMN, and N400. Here, we employ a series of explainable artificial intelligence (XAI) methods, including linear probing, masking, ablation, and activation maximization methods, to explore the nature of underlying representations and understand which features drive the latent representations of the GAE, as well as the dependence of the GAE on different elements in forming latent representations. This provides clearer theoretical support and human-understandable explanations for GAE-based weighting methods.

Contents

1 Introduction

Electroencephalography (EEG) is a non-invasive method that records the brain's electrical activity using electrodes placed on the scalp. Because EEG reflects the rapid postsynaptic potentials generated by a large number of neurons, it has millisecond-range temporal resolution and is widely used to study neuroscience, cognitive science, and motor processes Niedermeyer and Silva, 2005. Furthermore, EEG is the gold standard diagnostic procedure for epilepsy and sleep-related disorders. Event-related potentials (ERPs)—voltage fluctuations synchronized with stimuli—are extracted by analyzing multichannel EEG responses across repeated trials.

LIMO EEG is a toolbox for the statistical analysis of physiological data Pernet *et al.*, 2011. This toolbox introduces a hierarchical linear modeling (HLM/GLM) approach, which uses linear regression to model single-trial EEG responses at the subject level. These GLM parameters (a.k.a β coefficients) quantify neural activity under specific conditions at each channel and time point, represent the estimated contribution to the EEG signal, and serve as the basis for group-level statistical analyses such as t-tests and ANOVA. While the statistical method is powerful, it implicitly assumes consistent data quality across all channels and subjects, neglecting considerable variability caused by electrode reliability or individual neuroanatomical and cognitive differences.

To overcome these limitations, a Graph Autoencoder (GAE) has been introduced as a data-driven weighting method for second-level analysis (a.k.a group level). In this approach, a separate GAE model is trained for each β coefficient corresponding to a specific experimental condition or regressor. This design choice allows the model to capture the condition-specific spatiotemporal structure, rather than integrating all experimental effects into a shared latent representation. In this updated workflow, GAE-derived weights aim to reduce the weight of unreliable subjects/channels in 2nd-level analyses. Instead of relying on outlier detection methods that are blind to spatial-temporal dependencies, GAE-derived weights reflect deviations from common spatial-temporal activity. The GAE learned a latent representation for each β coefficient corresponding to each experimental condition or regressor in the GLM in the design matrix, and the differences between the GAE modelling and GLM modelling are used for weighting.

In addition, this thesis primarily focuses on the interpretability of the GAE. We employed a series of explainable artificial intelligence (XAI) techniques, including linear probing method; channel, region, and time frame masking methods; latent space, model module, graph edge, subject, and channel ablation methods; and activation maximization method to systematically discover what GAEs learn and how different elements contribute to forming their latent representations. These analyses distinguished the roles of different elements in GAE learning and evaluated whether the learned latent representations could capture meaningful EEG structures.

By integrating graph-based autoencoders with comprehensive xAI analyses, this work aims not only to introduce a GAE-based weighting strategy to enhance group-level analysis but also to provide an understanding of the latent space learned by GAEs to explain how data-driven weights are derived from EEG data.

2 Methods and Materials

This section introduces the datasets and event-related potential (ERP) paradigms used, also introduces the electroencephalogram (EEG) processing steps and the GAE model used to reconstruct β values. Then describes a series of explainable artificial intelligence (XAI) methods we used for evaluating the contribution of different elements and interpreting latent features, discovering and explaining how GAE processes EEG data and forms the latent representation. The explainable artificial intelligence (XAI) is the technique used to study and interpret AI models, allowing human users to understand and trust the outcomes produced by machine-learning systems (Adadi and Berrada, 2018).

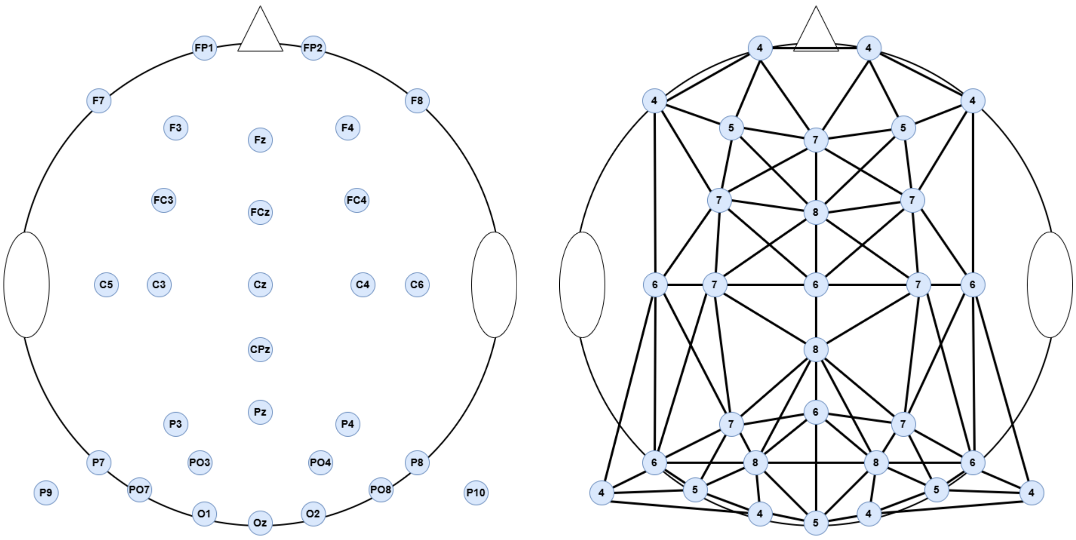


Figure 2.1.: Left: Scalp map showing the spatial layout and names of the 30 EEG electrodes in the International 10/20 System;
Right: The corresponding channel connectivity graph. Each node is labeled with its graph degree. Edges represent adjacency relationships derived from the channel adjacency matrix used in GAE.

Index	Channel	Degree	Neighbors	Index	Channel	Degree	Neighbors
1	FP1	4	F3, F7, FP2, Fz	16	FP2	4	FP1, Fz, F4, F8
2	F3	5	FP1, F7, FC3, Fz, FCz	17	Fz	7	FP1, F3, FC3, FP2, F4, FC4, FCz
3	F7	4	FP1, F3, FC3, C5	18	F4	5	FP2, Fz, F8, FC4, FCz
4	FC3	7	F3, F7, C3, C5, Fz, FCz, Cz	19	F8	4	FP2, F4, FC4, C6
5	C3	7	FC3, C5, P3, P7, CPz, FCz, Cz	20	FC4	7	Fz, F4, F8, FCz, Cz, C4, C6
6	C5	6	F7, FC3, C3, P3, P7, P9	21	FCz	8	F3, FC3, C3, Fz, F4, FC4, Cz, C4
7	P3	7	C3, C5, P7, PO7, PO3, Pz, CPz	22	Cz	6	FC3, C3, CPz, FC4, FCz, C4
8	P7	6	C3, C5, P3, P9, PO7, PO3	23	C4	7	CPz, FC4, FCz, Cz, C6, P4, P8
9	P9	4	C5, P7, PO7, O1	24	C6	6	F8, FC4, C4, P4, P8, P10
10	PO7	5	P3, P7, P9, PO3, O1	25	P4	7	Pz, CPz, C4, C6, P8, PO8, PO4
11	PO3	8	P3, P7, PO7, O1, Oz, Pz, CPz, PO4	26	P8	6	C4, C6, P4, P10, PO8, PO4
12	O1	4	P9, PO7, PO3, Oz	27	P10	4	C6, P8, PO8, O2
13	Oz	5	PO3, O1, Pz, PO4, O2	28	PO8	5	P4, P8, P10, PO4, O2
14	Pz	6	P3, PO3, Oz, CPz, P4, PO4	29	PO4	8	PO3, Oz, Pz, CPz, P4, P8, PO8, O2
15	CPz	8	C3, P3, PO3, Pz, Cz, C4, P4, PO4	30	O2	4	Oz, P10, PO8, PO4

Table 2.1.: Graph degree and neighboring channels for each EEG electrode based on the scalp adjacency matrix.
Note that the channel index used here corresponds to the order of the graph nodes in the adjacency matrix, which may differ from the default channel numbering in EEGLAB; the channel numbering in EEGLAB can be found in Table A.1 in the appendix A.

2.1 Dataset

The data used comes from ERP CORE (Kappenman *et al.*, 2021) (Stewart *et al.*, 2021) (Open Resources and Experiments Compilation), a free and standardized EEG resource designed to support reproducible research in cognitive neuroscience. ERP CORE contains optimized experimental paradigms, experimental control scripts, sample data from 40 participants, data processing workflows and analysis scripts, and extensive results for seven different ERP components obtained from six different ERP paradigms. This dataset provides high-quality, well-documented experimental protocols that reliably elicit typical ERP components with high signal-to-noise ratios. Recordings were obtained from 30 scalp electrodes mounted in an elastic cap and positioned according to the International 10/20 System. Figure 2.1 shows the location of the electrodes on the scalp and their adjacent relationships; The electrode layout and their adjacency structure exhibit a clear left–right hemispheric symmetry. Corresponding information can be found in Table 2.1.

The ERP CORE dataset has results for seven different ERP components obtained from six different ERP paradigms, but this study only focuses on four ERP components from four different ERP paradigms: MMN, N170, N400, and P3 (also known as P300). The study chooses these four paradigms because they have consistent and comparable design matrices across all subjects, enabling the generation of uniform β coefficient data for GAE training. In contrast, components such as N2PC, ERP and LRP involve lateralized designs that vary from individual to individual, making it impossible to directly align their β coefficient data for GAE-based weighted method.

The MMN is elicited by the Passive Auditory Oddball Paradigm that presents standard and deviant auditory tones, evoking a midfrontal negative stimulus of approximately 150–250 ms. The N170 is elicited by the Face Perception Paradigm that presents images of faces and objects, evoking a sharp occipitotemporal negative stimulus of approximately 170 ms. The N400 is elicited by the Word Pair Judgement Paradigm that uses semantically consistent and inconsistent sentence endings. The waveform peaks approximately 400 milliseconds after the stimulus appears, and its negative wave can be observed within a time window of 250 to 500 milliseconds. The P3 is elicited by the Active Visual Oddball Paradigm, which introduces an uncommon target stimulus within a standard stimulus sequence, evoking a central parietal positive stimulus of approximately 250–500 ms. Figure 2.2 shows the example trials of the four ERP paradigms. And the trial type and experimental conditions corresponding to the β index of all ERP components can be found in Table A.2, Table A.3, A.4 and Table A.5 in Appendix A.

2.2 EEG Processing

Each task is processed using a standardized BIDS-EEGLAB workflow. In short, the raw EEG data for each task were organized using the BIDS-EEG specification and imported into EEGLAB, which automatically generated a STUDY structure and unified metadata such as channel locations and event information. Then it will follow the fully automated pipeline recommended by Pernet *et al.* (2021): bad channels were detected and removed, the data were high-pass filtered at 0.5 Hz, re-referenced to the average, and artifacts were removed using ICLabel-based (Pion-Tonachini *et al.*, 2019) artifact component rejection. Remaining artifacts were identified and removed using the Artifact Subspace Reconstruction (ASR) method (Kothe and Jung, 2016). Clean continuous data were then epoched around specific task events, and first-level statistical modeling was performed using LIMO EEG to obtain single-subject parameter estimates, which were subsequently used for group-level analysis.

Following the LIMO EEG framework (Pernet *et al.*, 2011), at the single-subject level, for each electrode and at each time point, the EEG data of a single trial were modeled using a GLM. Let $\mathbf{Y}$ denote the trial-by-time data matrix of size $p \times n$, where p is the number of trials and n the number of time points. For each trial, the experimental operations are coded in a design matrix $\mathbf{X}$ of size $p \times m$, with m columns representing experimental conditions or covariates. The linear model is written as:

$$
\begin{aligned}
&\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}, \\
&\mathbf{Y} \in \mathbb{R}^{p\times n},\ \mathbf{X} \in \mathbb{R}^{p\times m},\ \mathbf{B} \in \mathbb{R}^{m\times n},\ \mathbf{E} \in \mathbb{R}^{p\times n}
\end{aligned}
\tag{2.1}
$$

where $\mathbf{B}$ is a $m \times n$ matrix of estimated regression parameters (β coefficients), and $\mathbf{E}$ is the residual error term. For each time point, the model reduces

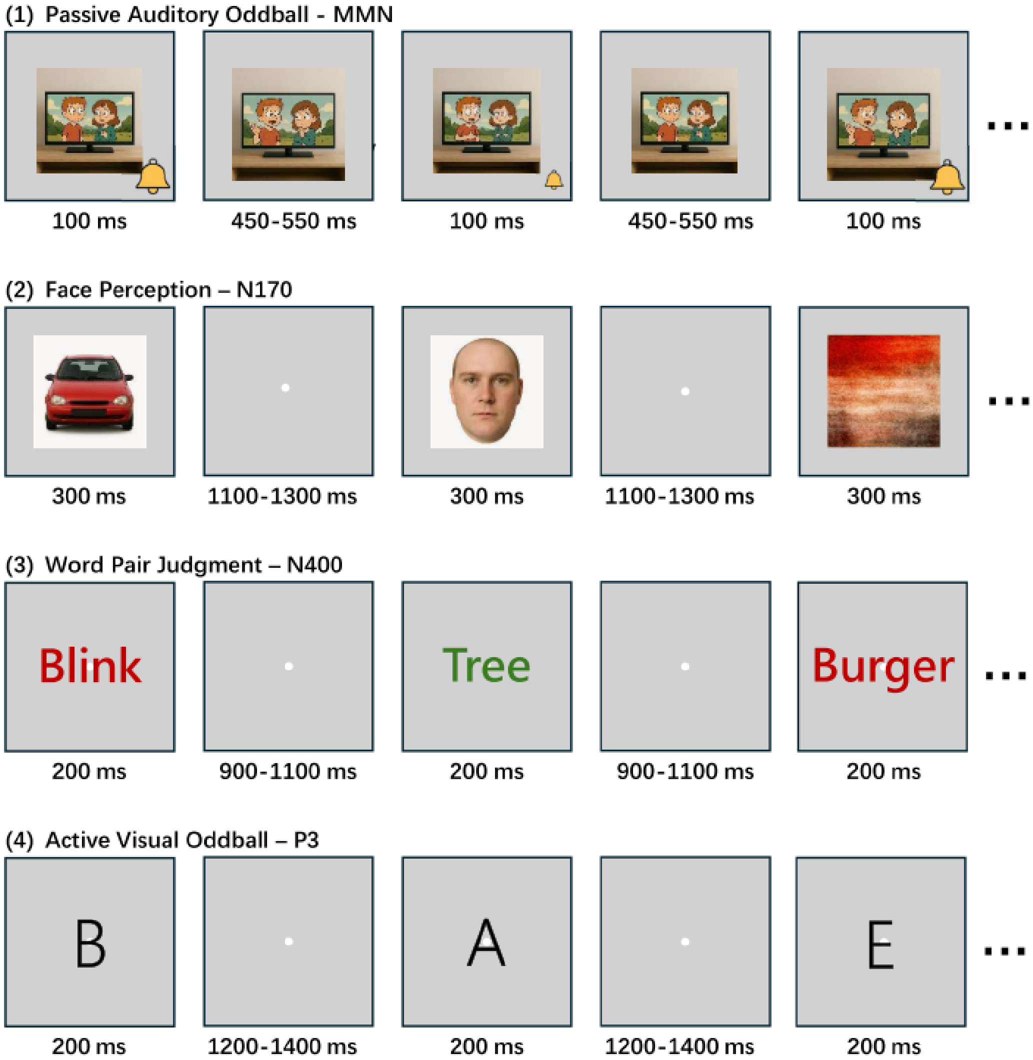


Figure 2.2.: Example of trails in each of the four paradigms. (1) Participants watched a silent video while standard and deviant audio was played to test whether the brain automatically detected the changes in sound. (2) Participants judged whether the image presented in the center of the screen was an "object" (face or car) or a "texture" (scrambled face or scrambled car). (3) In each trial, a pair of words, one red and one green, are presented, and participants judge whether the two words are semantically related. (4) In each trial block, five letters A–E are presented randomly with equal probability. The experiment designates one of these letters as the "target" of that block, and participants must determine whether it is the target. (Kappenman *et al.*, 2021)

to a least squares problem, the solution of which can be obtained using the generalized Moore-Penrose pseudoinverse:

$$\mathbf{B} = \left(X^{\mathsf{T}}X\right)^{-1} X^{\mathsf{T}}Y = \mathbf{X}^{+}\mathbf{Y}, \tag{2.2}$$

where $\mathbf{X}^{+}$ is the pseudoinverse of $\mathbf{X}$.

At the group level, effects of interest are assessed by contrasting β parameters across subjects using classical tests, including one-sample and paired t-tests, ANOVA, etc. Within the LIMO EEG framework, these tests can be conducted using the mean-based (a.k.a mean method) method or the trimmed mean-based (a.k.a robust method) method, thus increasing resistance to outliers. In addition to these established methods, we investigated a new option: subject-weighted statistics. In this strategy, each subject contributes differently at the group level, according to a data-driven reliability estimate (see below).

2.3 GAE Model

Graph autoencoders (GAEs) are unsupervised neural network architectures designed to learn low-dimensional representations of graph-structured data (Kipf and Welling, 2016). Unlike traditional autoencoders based on independent feature vectors, GAEs incorporate the connectivity structure of the input graph, allowing information to flow between neighboring nodes.

To learn the spatial information representation of multichannel EEG activity, the method for reconstructing β employs a GAE-based approach. In this framework, each EEG channel is treated as a node in a graph, with the anatomical and spatial adjacency relationships between electrodes defining the graph's edges. For each β matrix, the node feature vector corresponds to the temporal evolution of the channel response (n time points), thus generating an input of shape $[nChan \times nTime]$ for each subject. This representation allows the model to utilize the inherent spatial structure of EEG leads during encoding and reconstruction.

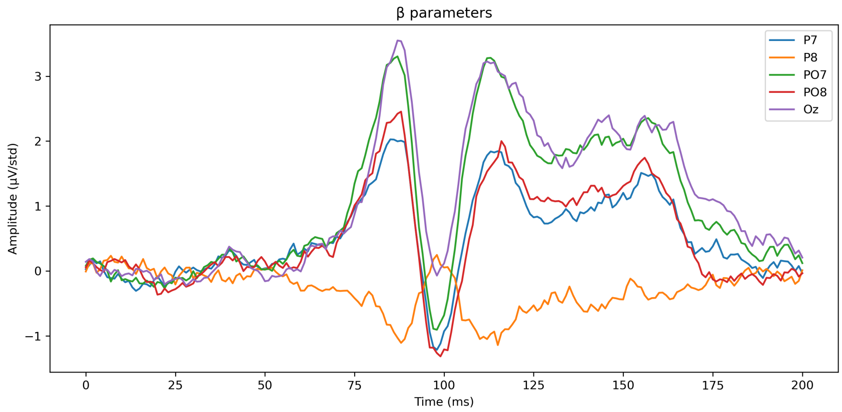


Figure 2.3.: Example of the input β parameters. Different colored lines represent β values for different channels.

The GAE encoder consists of two stacked AutoRegressive Moving Average (ARMA) graph convolutional layers (Bianchi *et al.*, 2021). ARMA graph convolution offers greater expressive power and stability compared to traditional graph convolutional network (GCN) layers. ARMAConv approximates a richer frequency response by combining multiple shared ARMA filters, each of which is equivalent to a recursive graph convolution. This approach avoids the oversmoothing problem common in deep GCNs.

The model is trained using a self-supervised approach to reconstruct the features of the input nodes. Given an input graph, its node feature matrix is $\mathbf{X}$ and its reconstruction $\hat{\mathbf{X}}$, the GAE parameters were optimized using the Smooth L1 (Huber) loss (Huber, 1992)

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{SmoothL1}\big(\hat{\mathbf{x}}_i, \mathbf{x}_i\big),$$

where for each element x and its prediction $\hat{x}$

$$\mathrm{SmoothL1}(\hat{x}, x) = \begin{cases} 0.5\,(\hat{x} - x)^2, & \text{if } |\hat{x} - x| < 1, \\ |\hat{x} - x| - 0.5, & \text{otherwise.} \end{cases}$$

For small errors, this loss exhibits a quadratic relationship; for large errors, it exhibits a linear relationship, thus maintaining a smooth gradient while being

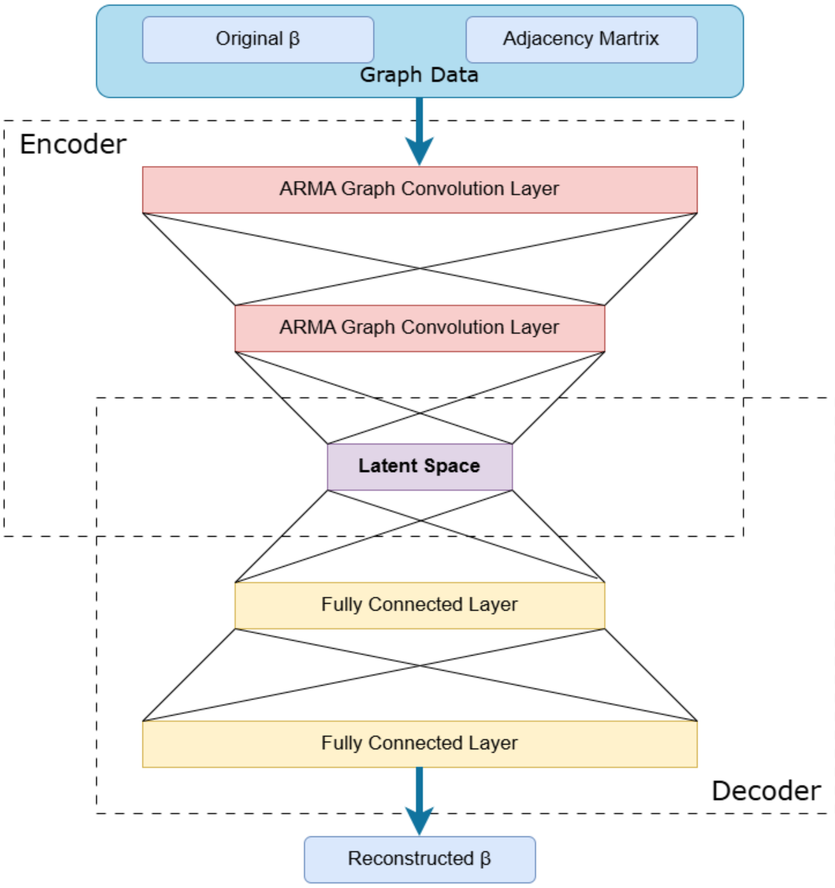


Figure 2.4.: Overview of GAE. The first ARMA convolutional layer projects the input feature dimension ($nTime$) into a 128-dimensional intermediate space, and the second ARMA convolutional layer maps the output to a 32-dimensional latent embedding. The decoder reconstructs the input signal using two fully connected layers, generating a reconstruction matrix of shape $[nChan \times nTime]$, which can be directly compared with the original input.

robust to outliers, which is ideal for noisy EEG data. In practice, we used the Adam optimizer with a learning rate of 10^{-3} and trained each GAE for 200 epochs with a mini-batch size of 4.

The trained GAEs are applied to each GLM parameter (β) to obtain the reconstructed β (hereinafter referred to as β_{GAE}). Similar to the standard GLM procedure, where the β coefficients are used together with the design matrix to compute the modelled data $\hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}}_{\text{GAE}}$. To quantify the reliability for each subject, we calculate the absolute difference between the original modelled data and its GAE-based modelled data. These differences can be globally averaged (across channels and time) to obtain a subject-level error metric. Then, weighting factors are derived by applying an inverse logarithmic transform to these error metrics, such that larger reconstruction differences correspond to smaller weights. In practice, errors are normalized using a z-score transformation, and weights are computed as the negative exponent of the normalized error. This process generates a weight for each subject, which is subsequently used in group-level statistical analyses.

2.4 Linear Probing

To evaluate the interpretability and information content of the latent representations learned by GAE, we employed linear probing analysis. Linear probing is a widely used technique in representation learning that evaluates the ability of a simple linear model to decode meaningful information from fixed latent embeddings (Alain and Bengio, 2016). Since the probing model itself is very simple, typically a linear classifier or regressor without hidden layers, good decoding performance indicates that the latent space contains sufficient information without relying on downstream layers of the model.

In this study, linear probing was applied to the latent embeddings generated by the encoder. After training on a given β component, we extracted the latent representation and then used the latent space as input features to train a simple linear decoder, which only contained a fully connected layer. During probing, the GAE parameters were frozen to ensure that the probing evaluated the quality of the learned latent space representation, rather than influencing it. The process is shown in Figure 2.5.

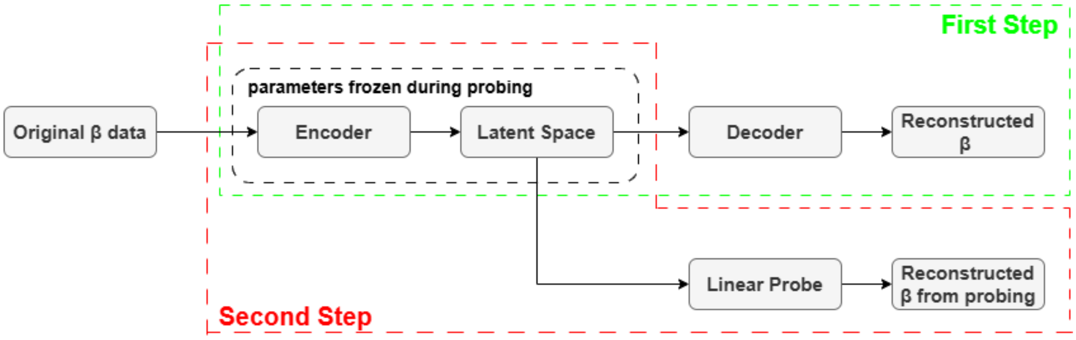


Figure 2.5.: Overview of Linear Probing process. Step 1: The GAE is trained end-to-end to reconstruct the β parameters. Step 2: The encoder is frozen, and a linear probe (a single fully connected layer) is trained on the latent representations to reconstruct β.

Finally, the results were evaluated by comparing the mean absolute error (MAE) of the probing result with the MAE of the original reconstruction. A higher decoding accuracy indicates that the GAE successfully encoded sufficient information within its latent dimension. Conversely, poorer performance suggests that the latent space failed to preserve information relevant to the β parameter features.

2.5 Masking

To evaluate the contribution of different channels, regions, and time frames of EEG to the GAE, we performed a series of masking analyses. Masking is an interpretable artificial intelligence technique, and we used Prediction Difference Analysis (Zintgraf *et al.*, 2017), which quantifies the impact of systematically masking specific input channels, brain regions, or time frames on the model's ability to learn representations, i.e., the model's dependence on specific features. The PDA's importance does not measure the reconstruction effect of specific features, but rather the indispensability of specific features to the internal representation of the model. Since the training objective of the GAE is to reconstruct a multi-channel beta matrix, the increase in reconstruction error after masking indicates that the masked components contain information crucial to the learned representation.

We employed three masking methods: channel masking, region masking, and time frame masking. Across all masking analyses, importance scores were computed as the absolute difference between the masked and original reconstruction error,

$$\mathrm{Imp}_{ch} = \left| \mathcal{L}_{\mathrm{SmoothL1}}(x_{\mathrm{masked},ch}, \hat{x}_{\mathrm{masked},ch}) - \mathcal{L}_{\mathrm{SmoothL1}}(x, \hat{x}) \right|,$$

using the Smooth L1 loss (Huber loss) as the reconstruction metric.

2.5.1 Channel Masking

Channel masking assesses the importance of individual electrodes by selectively masking one channel at a time. For each electrode, the original signal is replaced with zero or the mean of the remaining channels. The variation in reconstruction error relative to the unmasked baseline indicates the degree to which the GAE depends on the information of that channel.

2.5.2 Region Masking

The region masking method involves defining the region for each channel as itself and its neighborhood on the EEG adjacency map. The neighborhood structure is derived from the EEG adjacency matrix, where edges encode spatial and functional connectivity between electrodes. Specifically, channels are considered neighbors if they are directly connected in the adjacency matrix. The neighbor relationships can be seen in the Figure 2.1 and Table 2.1. Then, we perturb all channels within that neighborhood using one of three masking modes: zero masking, global mean masking, or local mean masking. The masked input signal is then passed through a GAE model, and the change in reconstruction error relative to the unmasked input signal is used as a region importance score. This method assesses the dependence of the GAE on each electrode and its neighboring electrodes during reconstruction.

2.5.3 Time Frame Masking

Time frame masking aims to quantify which time segments of the EEG time series are most informative for the GAE reconstruction. Since each EEG channel is represented as a graph node with a time-series feature vector, this analysis perturbs contiguous segments along the time dimension while keeping the trained model fixed, following the principle of PDA.

For each input sample $x \in \mathbb{R}^{nChan \times nTime}$, we uniformly partitioned the time axis into K non-overlapping segments ($K = n_{\text{segments}}$) using equally spaced boundaries,

$$b_k = \left\lfloor \frac{k}{K} nTime \right\rfloor, \quad k = 0, \ldots, K,$$

so that segment k corresponds to the interval $[b_k, b_{k+1})$. We then masked one temporal segment at a time across *all* channels while leaving the remaining time points unchanged.

Two masking modes were considered. In zero masking, all signal values of all channels within the selected time segment are set to zero. In mean masking, the signal values within the selected time segment are replaced with the average signal value of the same channel over the remaining unmasked time segment.

The importance of a time segment is defined as the absolute increase in reconstruction error relative to the unmasked baseline. The larger the increase, the more the corresponding time segment contains temporal information patterns that are crucial for the reconstruction of the β coefficient.

2.6 Ablation

To evaluate the contribution of different components of the GAE, including latent representations, graph structure, model modules, and input channels, we performed a series of ablation analyses. In contrast to masking analyses, which perturb the input data, ablation analyses assess component importance by functionally removing or disabling specific internal representations or structural elements of the trained model while keeping the input and other components unchanged. The importance of a component is quantified by the increase in reconstruction error induced by its removal (Morcos *et al.*, 2018).

In all ablation experiments, reconstruction performance was measured using the Smooth L1 loss, and importance scores are computed as the difference between the ablated and unablated reconstruction errors.

2.6.1 Latent Ablation

To investigate the role of each latent dimension in the learned latent representation, we performed latent dimension ablation on the trained GAE. For each batch of data, the latent representation is first obtained using the encoder. Then, the parameter of one latent dimension is selectively set to zero across all nodes before being passed to the decoder, while all other dimensions remain unchanged.

For each latent dimension, its reconstruction error is computed after ablation and compared to the baseline reconstruction error of the original model. The increase in reconstruction loss is used as a latent importance score. This analysis identifies which latent dimensions contain the most effective information for reconstructing the specific β value.

2.6.2 Channel Ablation

To evaluate the contribution of each channel to the reconstruction, we conducted channel ablation experiments, analyzing their signal content and their role in the graph structure. For each channel, the corresponding node is removed from the graph along with all edges connected to it, and the associated node features are excluded from the input. We then fed the resulting reduced graph and reduced data into the trained GAE.

This process differs from channel masking. While channel masking perturbs the input signal of a channel but preserves the graph topology and the presence of the node, channel ablation completely removes the channel from the graph, eliminating both its signal information and its role in message passing on the graph.

We compared the reconstruction error of the ablated input to the baseline reconstruction error. Baseline reconstruction error is the reconstruction error of the original model on channels other than the ablated one. A greater increase in reconstruction loss indicates that the removed channel, together with its associated graph connections, played a more significant role in the graph-based reconstruction process.

2.6.3 Edge Ablation

To examine the dependence of the GAE on the spatial structure of the graph, two different edge ablation strategies are employed: random edge ablation and node-based edge ablation.

In the random edge ablation analysis, a fixed proportion of edges is randomly removed from the graph. The modified graph is used to reconstruct the input data, and the relative increase in reconstruction error was measured. This method evaluates the model's robustness to perturbations in global graph connectivity and measures whether the applied graph structure is effectively utilized during reconstruction, rather than behaving as a channel-wise autoencoder.

In the node-based edge ablation analysis, all edges connected to a specific channel were removed, effectively isolating that node from the rest of the graph while preserving its signal features. The resulting increase in reconstruction loss reflects the importance of the spatial relationships between that channel and its adjacent electrodes.

2.6.4 Module Ablation

To evaluate the contribution of each module, the module ablation is performed by perturbing the intermediate feature representations of selected encoder and decoder modules. Specifically, a forward hook was applied to the target module, and randomly setting a fixed proportion of its parameters to zero. The same random mask is applied consistently across all samples within an evaluation run.

The reconstruction error computed after module ablation is compared to the baseline error. The significant increase in loss indicates that the corresponding module plays a crucial role in GAE and supports accurate reconstruction.

2.6.5 Subject Ablation

To evaluate the contribution of individual subjects to the GAE, we performed a subject ablation analysis based on retraining. Since the GAE model is constructed using concatenated data from all subjects and trained separately for each β component, each subject constitutes part of the training distribution, rather than an input-level feature. Therefore, removing a subject modifies the distribution used to learn latent representations. Unlike input-level masking analysis (which perturbs the input while keeping the model fixed), subject ablation necessarily requires retraining the model to assess how excluding a subject affects the formation of the latent space during optimization.

For each subject s, we first evaluated the reconstruction performance of the original model trained on the raw dataset. Two baseline reconstruction errors were computed: (i) the reconstruction loss of subject s itself (*self*), and (ii) the average reconstruction loss of all remaining subjects (*others*). These baseline losses show how well the model reconstructs the excluded subject and the rest of the dataset, respectively.

Next, we retrained the GAE after removing subject s from the training dataset, while keeping the model architecture, training procedure, and hyperparameters unchanged. Using the retrained model, we again computed the reconstruction loss for the subject s and the mean reconstruction loss for the other subjects. The importance of the subject is quantified as the change in reconstruction error induced by retraining without that subject, separately for the *self* and *others* cases.

The change in self-reconstruction loss reflects how strongly the latent representation learned from other subjects generalizes to the excluded subject. In contrast, the change in reconstruction loss for other subjects indicates how much subject s contributes to shaping the GAE. A larger increase in reconstruction error for the other subjects indicates that the excluded subject provided informative, distinctive patterns that were important for learning the β.

This analysis was conducted exclusively on the ERP CORE dataset and is intended to evaluate subject contributions within this specific dataset.

2.7 Activation Maximization

To further interpret what the trained GAE encodes, we applied activation maximization (AM), an optimization-based xAI technique that synthesizes an input pattern that maximizes the activation of selected latent representations while keeping model parameters fixed (Erhan *et al.*, 2009) (Yosinski *et al.*, 2015). In our setup, AM is used to generate channel-temporal prototypes that strongly activate the latent representations of the encoder, thereby revealing the spatiotemporal patterns favored by the learned representations.

The following procedure is adopted to perform activation maximization. Let $x \in \mathbb{R}^{N_{\text{chan}} \times N_{\text{time}}}$ denote a channel–time input matrix and let $Z = f_{\text{en}}(x, A)$ be the latent representation generated by the encoder with the adjacency matrix A. Two activation maximization modes were considered: a global latent mode that maximizes the average activation of all latent dimensions to capture overall representation preferences; and a dimension-specific mode that maximizes the activation of each latent dimension to represent dimension-wise preferences. Therefore, the activation value to be maximized is defined as the average activation value across all latent dimensions,

$$\mathrm{Act}(x) = \frac{1}{N_{\text{chan}} D} \sum_{c=1}^{N_{\text{chan}}} \sum_{d=1}^{D} Z_{c,d},$$

or the mean activation of a specific latent dimension d,

$$\mathrm{Act}_d(x) = \frac{1}{N_{\text{chan}}} \sum_{c=1}^{N_{\text{chan}}} Z_{c,d},$$

where D is the number of latent dimensions. The input was optimized by minimizing the following objective function,

$$\mathcal{J}(x) = -\mathrm{Act}(x) + \lambda_2 \, \mathbb{E}[x^2] + \lambda_{\text{TV}} \, \mathrm{TV}(x),$$

where the regularization terms constrain the structure of the input. The ℓ_2 regularization term is defined as

$$\mathbb{E}[x^2] = \frac{1}{N_{\text{chan}} N_{\text{time}}} \sum_{c=1}^{N_{\text{chan}}} \sum_{t=1}^{N_{\text{time}}} x_{c,t}^2,$$

which limits the amplitude of large signals and prevents optimization from producing unrealistically high input amplitudes. The total variation (TV) regularization is defined as

$$\mathrm{TV}(x) = \mathbb{E}_{c,t}[\,|x_{c+1,t} - x_{c,t}|\,] + \mathbb{E}_{c,t}[\,|x_{c,t+1} - x_{c,t}|\,]$$

which encourages smoothness. The optimization process uses the Adam optimizer with a fixed number of iterations and randomized initial values. The optimized input serves as an AM prototype, revealing the spatiotemporal EEG patterns preferred by the learned latent representations.

Since the optimized AM prototypes are waveforms, direct visualization is often unable to reveal interpretable structures (e.g., consistent ERP waveforms) across different latent dimensions or experimental conditions, unlike images, which can be directly observed with the eye. To investigate the learned patterns, we further quantified the similarity structure among AM prototypes using unsupervised clustering. Specifically, we explored: (i) whether latent dimension AM patterns exhibit clustering structures under different experimental conditions (i.e., different GLM β coefficients); and (ii) whether AM patterns from different channels form significant clusters.

2.8 Implementation

LIMO EEG functions have been expanded to support exporting the GAE parameters and custom Python functions were developed for the different aforementioned methods. This implementation allows users to apply Graph-based Autoencoder learning and explainable artificial intelligence (xAI) techniques to better understand the contributions of EEG features, graph structure, model components, and dataset characteristics to the latent representations and reconstruction performance.

The expanded workflow supports three operating modes. In the first mode, no intermediate information related to the GAE is saved. The GAE is used only to reconstruct the β values, and only the reconstructed β values are returned to the LIMO EEG framework.

In the second mode, the GAE model parameters corresponding to each stimulus-specific β component are saved, together with the original and reconstructed β data. In addition, channel location information and the EEG adjacency matrix defining the graph structure are stored.

The third mode performs all the above-described xAI analyses. In this mode, all the data mentioned in the previous mode, along with the raw results and visualizations obtained through the xAI procedure, are saved together.

Due to the substantial computational cost associated with retraining models, subject-level ablation is not performed within the implementation framework. Subject ablation requires retraining the GAE after removing individual subjects from the training set and is therefore prohibitively time-consuming for routine use. Consequently, subject ablation analyses were conducted only in dedicated experimental settings and are not included as a standard operating mode of the toolbox. Because retraining the model requires significant computational resources, the subject ablation analysis is only performed in an experimental setting and is not used as a standard operating procedure in the toolbox.

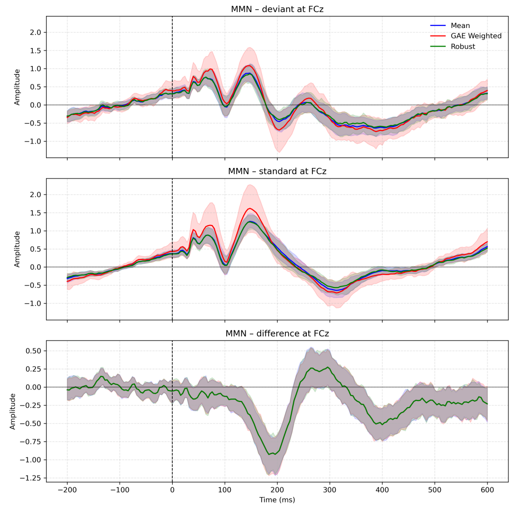
3 Results

3.1 GAE Model

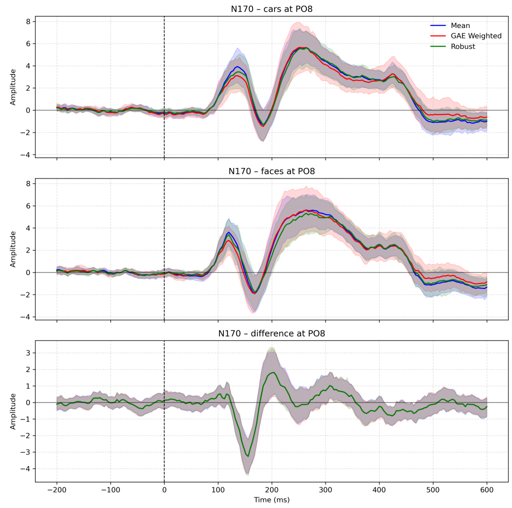
In the 2nd-level analysis of LIMO EEG, for each method, including classical mean statistics, trimmed means, and GAE-weighted means, we successfully reproduced the previously published effects (Kappenman *et al.*, 2021) on all ERP components, as shown in Figure 3.1 and Figure 3.2.

Figure 3.1 shows the ERP waveforms and their differences on representative electrodes under two different experimental conditions derived from three different methods: Mean, Robust and GAE-Weighted. In the MMN result, the waveforms of 'Deviant' condition and 'Standard' condition show a clear separation between 150–250 ms, with the difference waveforms exhibiting a distinct negative peak within this time window. In the N170 result, the 'Face' condition exhibits stronger negative fluctuations around 170 ms, while the 'Car' condition also has, but weaker, resulting in a significant negative peak in the difference waveform at around 160–190 ms. In the N400 result, the 'Unrelated' condition shows a stronger negative wave between 300–500 ms, while the wave of the 'Related' condition is relatively weaker within this time window, resulting in a negative peak in the difference waveform within around 300–500 ms. In the P3 result, the waveform of the 'Target' condition shows a significant positive wave after 300 ms, while the wave of the 'Distractor (Non-target)' is weaker, resulting in a positive peak in the difference waveform within around 350–600 ms. The result of all four ERP components exhibits typical ERP waveforms consistent with previously published effects in their representative channels.

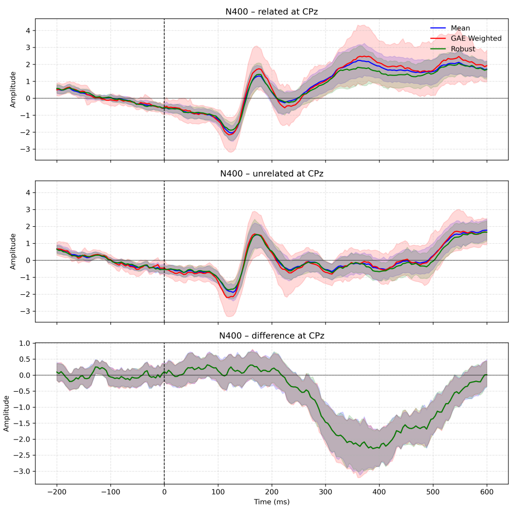
For all ERP components, all three methods produced highly similar ERP waveforms and their difference waveforms, indicating that the GAE-based weighting method preserves the typical structure and information of each ERP. However, we also found that the confidence space obtained by the GAE-based weighting



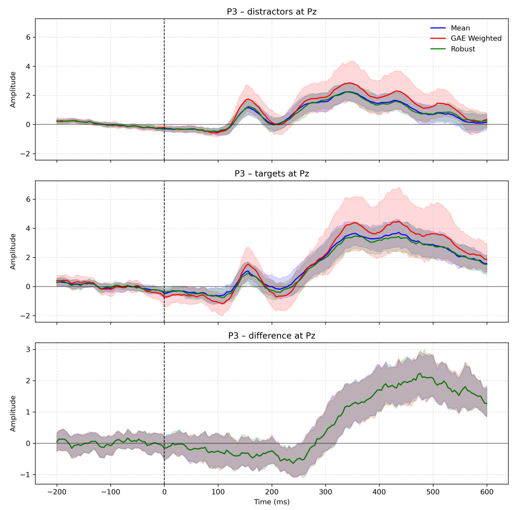
(a) MMN



(b) N170



(c) N400



(d) P3

Figure 3.1.: Comparison of ERP waveforms of three methods: Mean, Robust, GAE weighted for four ERP components. For each component, the two figures above show the ERP waveforms on its representative electrodes under two different experimental conditions (e.g., Deviant vs. Standard for MMN; Cars vs. Faces for N170; Related vs. Unrelated for N400; Distractors (Non-targets) vs. Targets for P3), while the figure below shows the waveforms showing the differences between the waveforms of two experimental conditions.

method is larger than that of the other two methods. This may be because the weights obtained by GAE increase the weight of well-reconstructed subjects and decrease the weight of poorly reconstructed subjects, making the final group estimate more dependent on certain subjects, thus increasing the confidence interval. In summary, the larger confidence interval reflects the redistribution effect among subjects, indicating that the weighting method is working, while the consistency among the waveforms suggests that it only changes the statistical structure rather than adding noise.

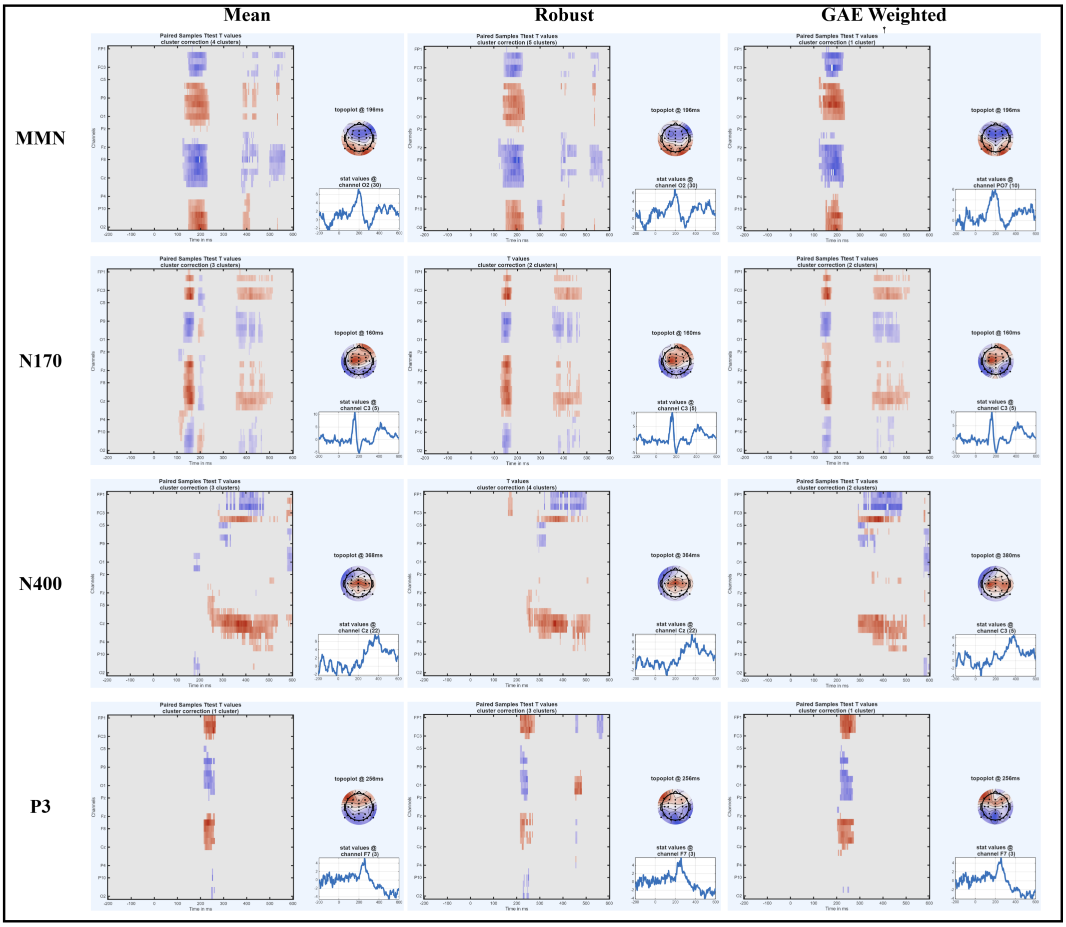


Figure 3.2.: Comparison of cluster-corrected paired-sample t-value maps across three methods for four ERP components. Paired-sample t-values are shown as channel-by-time maps for MMN, N170, N400, and P3, computed using three different methods: Mean (left column), Robust (middle column), and GAE-weighted (right column). Cluster-based correction (P=0.05) is used, so only samples that form significant clusters are visualized. The image in the upper right corner shows the scalp topography, and the bottom right graph shows the t-value time variation for representative channels.

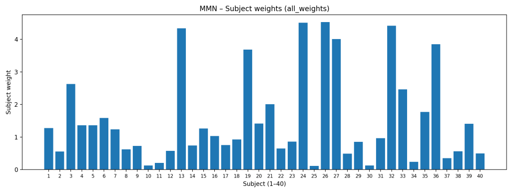
Figure 3.2 compares the spatiotemporal distribution of paired-sample t-values obtained using the mean method, the robust method, and the GAE-weighted method for four ERP components. The results show that MMN, N170, N400, and P3 all have significant clusters, consistent with published ERP results. Except for the weighted method, which alters the significance channels in MMN, the three methods show the same significance channels in the other paradigms. In terms of cluster number, the weighted method generally has the fewest clusters, while the robust method generally has the most.

	Mean				Robust				Weighted			
ERP component	Min	Max	Mean	Median	Min	Max	Mean	Median	Min	Max	Mean	Median
MMN	-1.2659	1.670	-0.0079	-0.0039	-1.7536	1.4141	-0.0153	-0.0148	-0.9740	0.9607	-0.0075	-0.0082
N170	-1.437	1.7561	0.0135	0.0017	-1.626	2.0726	0.0232	0.0141	-1.1305	1.6226	0.0163	0.0028
N400	-1.0930	1.2254	0.0179	0.0123	-1.2917	1.6824	0.0259	0.0205	-1.0881	1.0841	0.0078	0.0064
P3	-0.7680	0.8761	-0.0012	-0.0049	-1.4121	1.4444	0.0024	-0.0021	-0.6861	0.8758	-0.0005	-0.0055

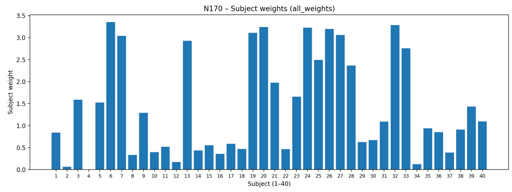
Table 3.1.: The effect size (Cohen's d) statistics for each ERP component under three group-level methods (mean, robust, and weighted). For each component and method, the table reports the minimum, maximum, mean, and median of Cohen's d across all channel-time samples.

Table 3.1 shows the effect size (Cohen's d) of each ERP component and method across the entire channel-time space. It can be observed that the mean and median of all Cohen's d are close to zero, reflecting the spatial and temporal locality of the ERP effect. This is consistent with the results we observed in Figure 3.2. Meanwhile, we can observe that the robust methods always generate the largest peak effect size across all ERP components, indicating their greater sensitivity to local effects, while the weighted methods generate more conservative extreme values. These results show that the GAE does not produce fake effects, but rather suppresses isolated, potentially noise-driven peaks, and this weighted method tends to provide smoother, more consistent results.

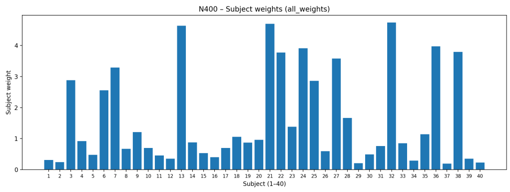
From Figure 3.3, we can observe that the weights of a subject vary in each ERP component, and the weight of each subject also differs across different ERP components. For example, in N400 and P3, Sub-21 and Sub-22 have high weights, while in MMN and N170, these two subjects have relatively low weights. Meanwhile, some subjects have high weights in all ERP components, such as Sub-13 and Sub-32, while others have low weights in all ERP Cs, such as Sub-02 and Sub-34. These observations indicate that participants' contributions to the group-level ERP analysis are component-specific. Instead of assigning fixed weights to participants, GAE-based methods adaptively adjust the contributions of different participants based on the learned representation. This demonstrates that the GAE-based weighting method learns the differences



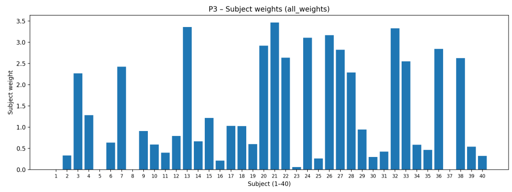
(a) MMN



(b) N170



(c) N400



(d) P3

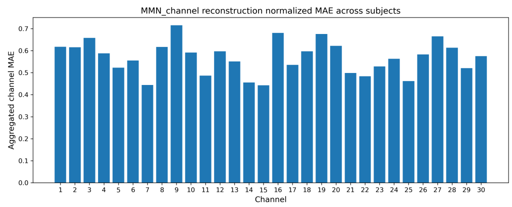
Figure 3.3.: Subject-level weights for 2nd-level analysis derived from the GAE-based weighting method for four ERP components. Note that due to data availability and design limitations, not all subjects' data participated in the N170 (missing sub-04) and P3 (missing sub-01, sub-05, sub-08, and sub-37) analyses; therefore, the weight of missing subjects is 0.

among participants and distinguishes their data quality, thus providing a basis for 2nd-level analysis.

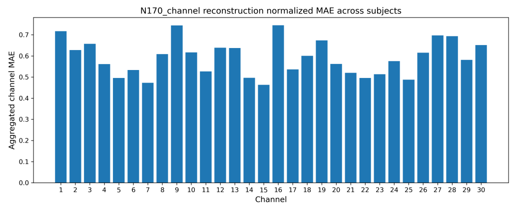
From Figure 3.4, it can be observed that the reconstruction errors for channels 16 (FP2), 1 (FP1), 9 (P9), 19 (F8), and 27 (P10) are relatively large across all ERP components. These channels are mostly located at the edge of the scalp, and these areas are more susceptible to individual differences and non-task activities. Conversely, channels 7 (P3), 14 (Pz), 15 (CPz), and 25 (P4) have relatively small reconstruction errors in all ERP components. These channels are mostly located in the parietal lobe and midline area.

3.2 Linear Probing

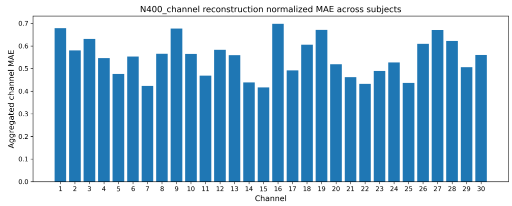
Table 3.2 lists the reconstruction errors of the original GAE and the linear probe analysis applied to latent representation learning for the four ERP components. For most β coefficients and ERP components, the average MAE of the probe is basically the same as the average MAE of the original GAE; for someβ coefficients, it is only slightly higher.



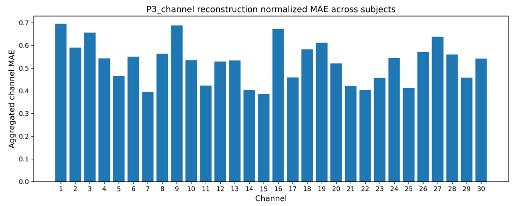
(a) MMN



(b) N170



(c) N400



(d) P3

Figure 3.4.: Channel-level reconstruction error across subjects for four ERP components. The higher the value in the figure, the greater the reconstruction error of that channel. The calculation method is to normalize the MAE value of each subject, and finally average the results across all subjects.

ERP Component	N400								N170				MMN	
Beta index	1	2	3	4	5	6	7	8	1	2	3	4	1	2
Original mean MAE	0.4159	0.4246	0.4291	0.4304	0.3973	0.3820	0.3969	0.3883	0.2433	0.2556	0.2311	0.2728	0.1006	0.0760
Probe mean MAE	0.4403	0.4352	0.4509	0.4460	0.4215	0.4016	0.4232	0.4117	0.2488	0.2594	0.2427	0.2787	0.1059	0.0788

ERP Component	P3												
Beta index	1	2	3	4	5	6	7	8	9	10	11	12	13
Original mean MAE	0.7937	0.8063	0.8415	0.8384	0.8095	0.7543	0.7672	0.7774	0.7863	0.7674	0.7515	0.7724	0.7832
Probe mean MAE	0.8183	0.7703	0.8495	0.7787	0.8533	0.8199	0.8666	0.8172	0.7777	0.8398	0.8798	0.7996	0.2484

ERP Component	P3											
Beta index	14	15	16	17	18	19	20	21	22	23	24	25
Original mean MAE	0.8093	0.7441	0.8250	0.7668	0.8294	0.7912	0.8470	0.7884	0.7637	0.8053	0.8386	0.8017
Probe mean MAE	0.8183	0.7703	0.8495	0.7787	0.8533	0.8199	0.8666	0.8172	0.7777	0.8398	0.8798	0.7996

Table 3.2.: Mean reconstruction error for original and linear probe reconstructions of four ERP components.

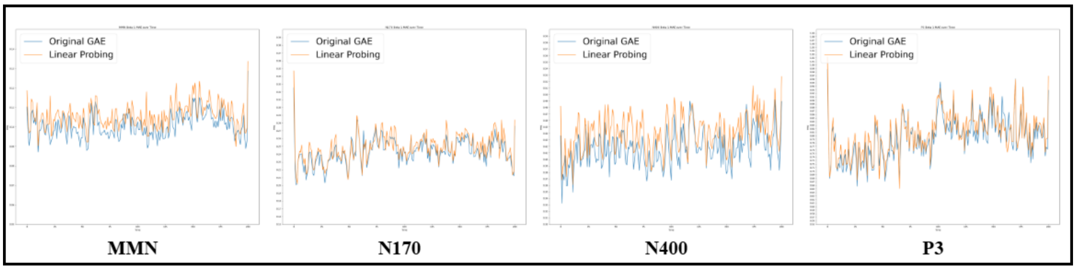


Figure 3.5.: Example reconstruction error temporal waveforms of the first β coefficient for four ERP components using the original GAE and linear probing.

Figure 3.5 shows reconstruction error temporal waveforms of the first β coefficient for MMN, N170, N400, and P3. The details of the experimental conditions of the relevant β coefficients are listed in Appendix A. It can be observed that the reconstruction error temporal waveforms generated by the linear probe are very consistent with the pattern of the original GAE. And this similarity can be found in all β coefficients of all ERP components.

The results indicate that most of the information needed to reconstruct the β coefficient data is preserved in the learned latent space and can be recovered using a simple linear mapping. This demonstrates that the latent representations learned by the graph autoencoder capture meaningful structures from the β data. It also indicates that existing GAE model designs can effectively learn the features of these four components.

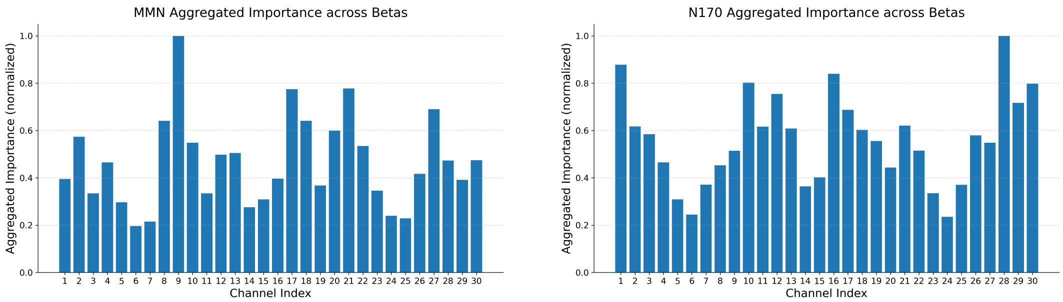
3.3 Masking

3.3.1 Channel Masking

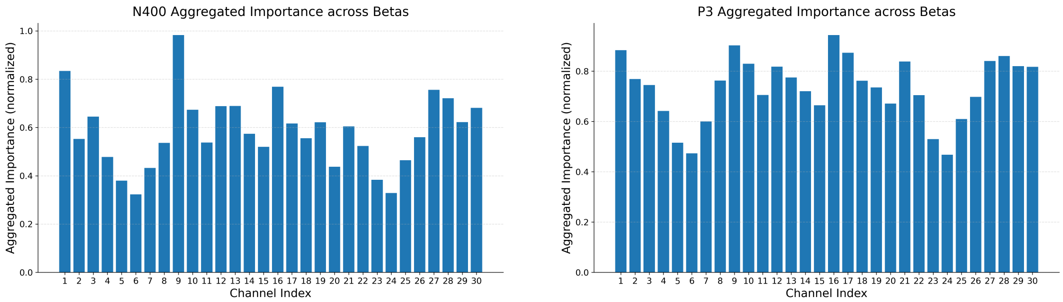
Figure 3.6 shows the aggregated channel importance for MMN, N170, N400, and P3 using the mean masking method. From the figure, we can observe that the channel importance distribution patterns of different ERP components are not entirely the same but differ, indicating that GAE has different dependencies on each channel when learning different components.

However, the results also indicate that the model relies more heavily on some certain channels, while the importance of some channels remains relatively low across components. Specifically, channels 1 (FP1), 9 (P9), 16 (FP2), and 17 (Fz) consistently have higher aggregated importance, indicating a stronger model reliance. In contrast, channels such as 6 (C5), 7 (P3), 24 (C6), 25 (P4) show lower importance values for all ERP components, suggesting a weaker overall influence on the model's learned latent representation.

In addition, we observed that for each ERP component, the channel importance of each β coefficient is not entirely the same, but their spatial distributions are highly similar, as shown in the example Figure 3.7, even compared with different ERPs (Figure 3.8) shows a certain degree of similarity. These similari-

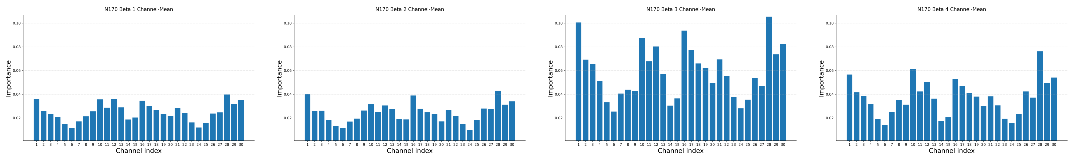


(a) MMN **(b)** N170



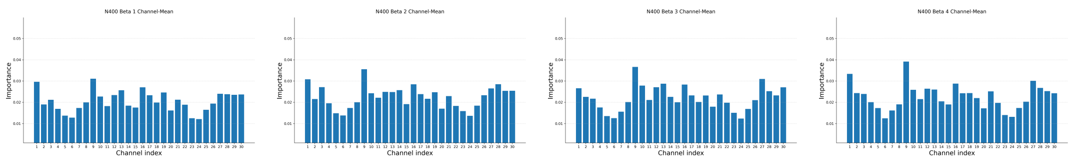
(c) N400 **(d)** P3

Figure 3.6.: Channel importance derived from channel masking for four ERP components. For each ERP component, importance scores were first normalized within each β coefficient and then averaged across all β.



(a) Cars **(b)** Faces **(c)** Cars-scrambled **(d)** Faces-scrambled

Figure 3.7.: Channel importance for the four β coefficients of N170. The detail of β coefficients is shown in Table A.3.

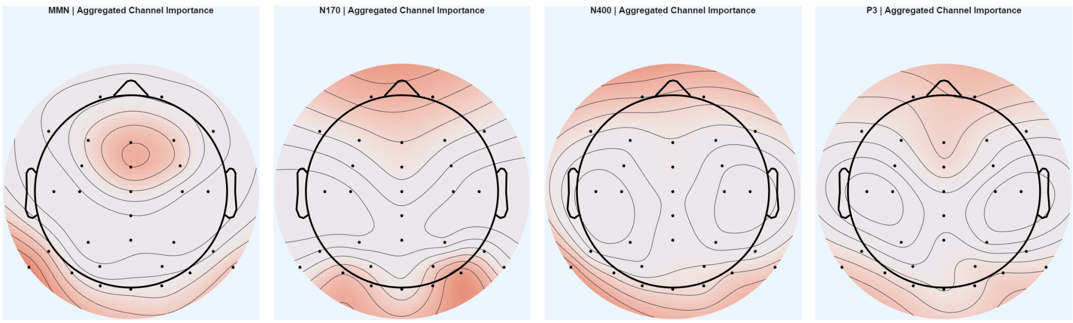


(a) Prime,related,list1 **(b)** Prime,related,list1 **(c)** Prime,unrelated,list1 **(d)** Prime,unrelated,list2

Figure 3.8.: Channel importance for the first four β coefficients of N400. The detail of β coefficients is shown in Table A.4.

ties indicate that GAE captures a stable hidden organizational structure that is consistently expressed under different experimental conditions and ERPs.

Comparing with the reconstruction errors of each channel in Figure 3.4, we can see that the distribution of reconstruction errors and importance results of channel masking varies greatly. Taking the N170 results as an example, some channels have large reconstruction errors and high importance, such as channels 1 (FP1), 16 (FP2), and 28 (PO8). This indicates that although these channels are difficult to reconstruct, the model's latent space learning relies on the unique information they provide. Conversely, some channels have small reconstruction errors with low importance, such as channels 6 (C5) and 7 (P3). We can consider these channels as redundant, as their signals are highly correlated with neighboring channels and without complex noise. Furthermore, some channels have large reconstruction errors but low importance, such as channel 9 (P9). This indicates that these channels may contain complex noise, and the model can not obtain sufficient information, also making reconstruction more difficult.

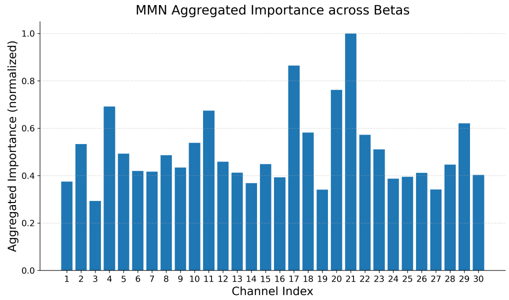


(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

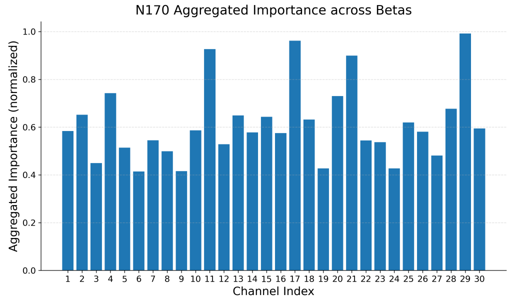
Figure 3.9.: Scalp topography derived from the channel masking results for the four ERP components. Warmer colors indicate that the channels are more important, reflecting a greater model dependence on these electrodes.

Figure 3.9 shows the distribution of channel importance on the scalp. Comparing it with the t-value distribution on the scalp in Figure 3.2, we find that for all ERPs, the distribution of important channels does not overlap significantly with the distribution of significant ERP effects. Based on our previous analysis, this indicates that the generation of the latent space of GAE relies not only on channels with significant ERP effects but also on some noisy channels that may contain key structural or important information.

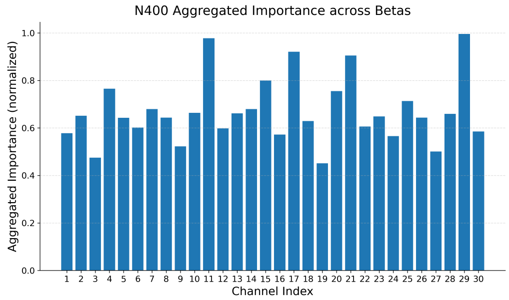
3.3.2 Region Masking



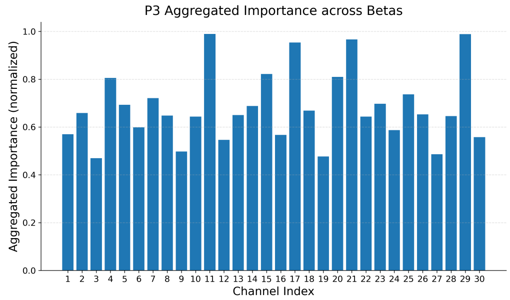
(a) MMN



(b) N170



(c) N400

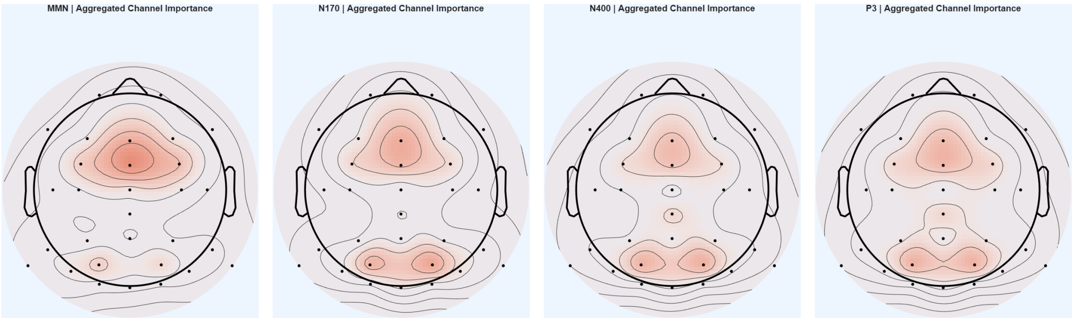


(d) P3

Figure 3.10.: Region importance (global) derived from region masking using the global mean method. For each ERP component, importance scores are first normalized within each β coefficient and then averaged across all β.

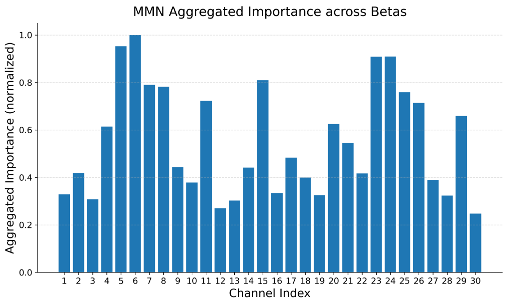
Figures 3.10 and 3.12 show the channel importance obtained from region masking using global and local mean strategies, respectively. Both methods show uneven channel importance across all ERP components, indicating that different channel and their neighbours (regions) contribute differently to the reconstruction performance. The comparison of Figures 3.11 and 3.13 reveals that the distributions of important regions obtained by the two methods differ. The important regions obtained using the global mean method are primarily distributed along the midline of the parietal and frontal lobes, while those obtained using the local mean method are mainly distributed in the parietal and temporal lobes. Both scalp topographies show a symmetrical distribution along the midline between the left and right hemispheres. These symmetrical distributions may be due to the symmetry of neural activity and the symmetry of node adjacency relationships (as shown in Figure 2.1).

Comparing the results of channel masking and region masking reveals a significant difference, which is consistent with our expectations and prior analyses:

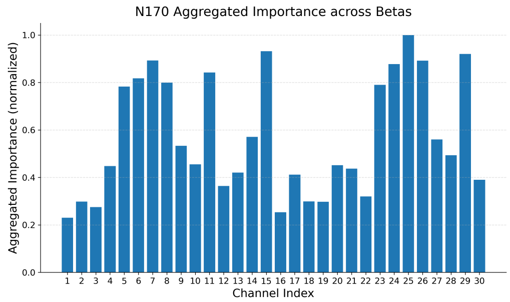


(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

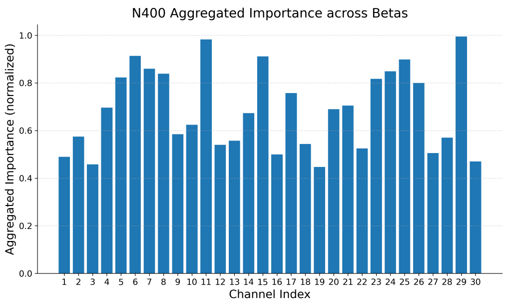
Figure 3.11.: Scalp topography (global) of region masking results using the global method for four ERP components.



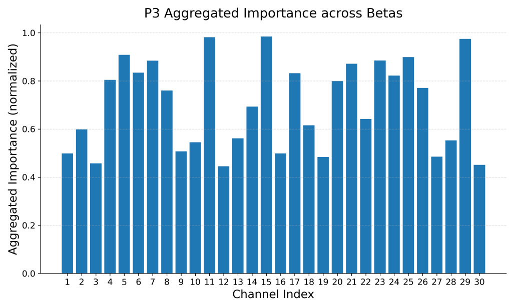
(a) MMN



(b) N170

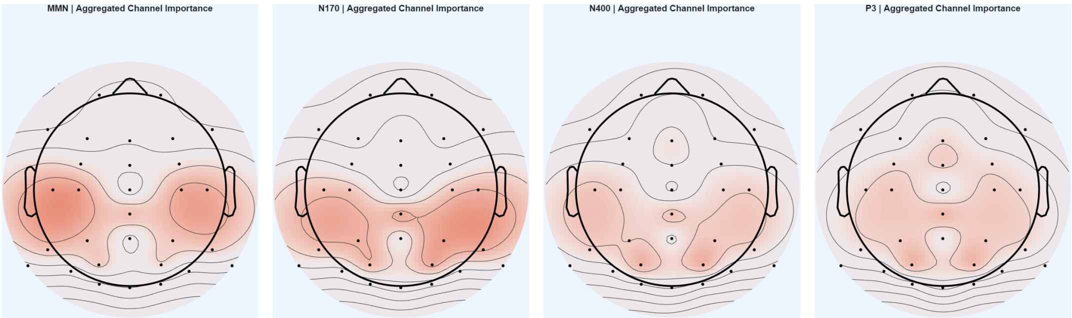


(c) N400



(d) P3

Figure 3.12.: Channel importance (local) derived from region masking using the local mean method. For each ERP component, importance scores were first normalized within each β coefficient and then averaged across all β.



(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

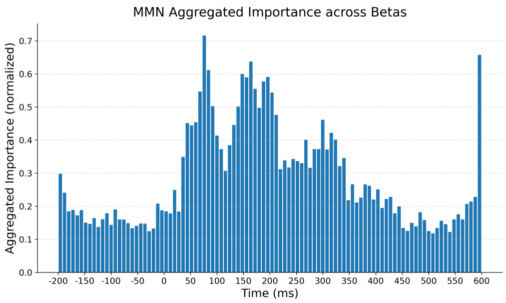
Figure 3.13.: Scalp topography (local) of region masking results using the local method for four ERP components.

First, the adjacency matrix indicates that nodes located in regions identified as important by region masking have high degrees in the graph 2.1, thus masking these regions effectively disables multiple highly connected nodes. Second, region masking explores the contribution of a region rather than the contribution of a single channel, and the regions identified as important contain rich information on which the model strongly relies.

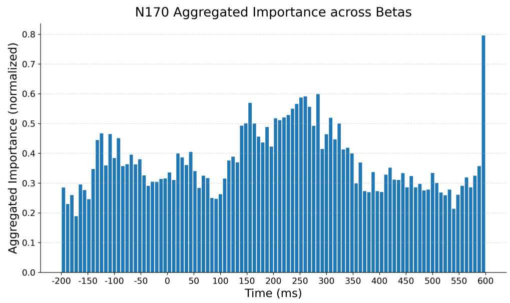
3.3.3 Time Frame Masking

Figure 3.14 shows the importance of different time frames obtained from the time-frame masking analysis. As shown in the figure, for the MMN component, the peak of importance is concentrated between 50 and 200 ms. For the N170 component, importance peaks primarily between 150 and 300 ms. In the case of N400, the highest importance is observed between 100 and 200 ms. The uneven distribution of the importance of time frames in some ERP components indicates that different time frame has varying impacts on reconstruction performance. And for the P3 component, the importance is distributed uniformly across time, indicating that contributions from different time points are relatively the same. Comparing Figure 3.1 and the time window of each ERP effect, the time importance patterns of MMN and N170 are partly consistent with their expected time windows, while N400 and P3 show a less direct correlation.

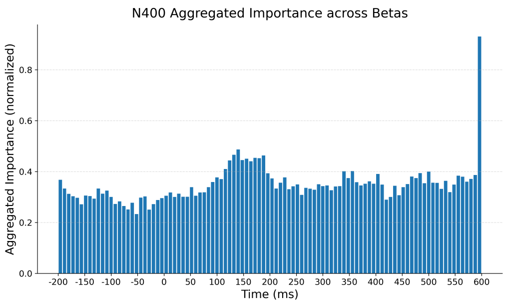
Furthermore, for each ERP component, the temporal importance distributions differ across individual β coefficients. Taking N170 as an example, as shown in Figure 3.15, under the 'cars' and 'faces' conditions, the peak of time importance



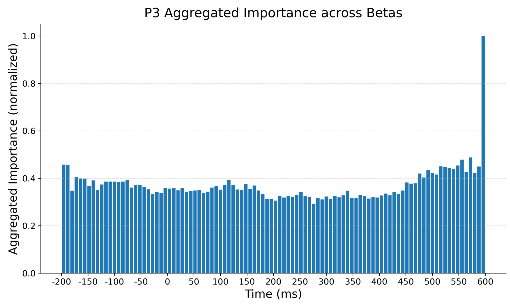
(a) MMN



(b) N170

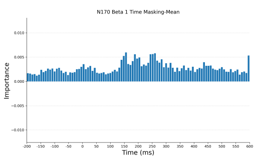


(c) N400

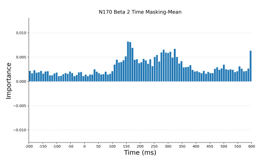


(d) P3

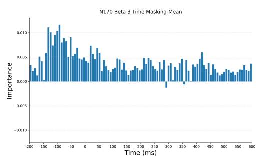
Figure 3.14.: Time-frame importance derived from time-frame masking across β coefficients, segment number $k = 100$. For each ERP component, importance scores were first normalized within each β coefficient and then averaged across all β.



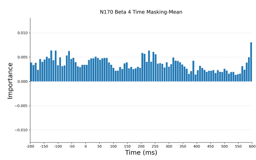
(a) Cars



(b) Faces



(c) Cars-scrambled



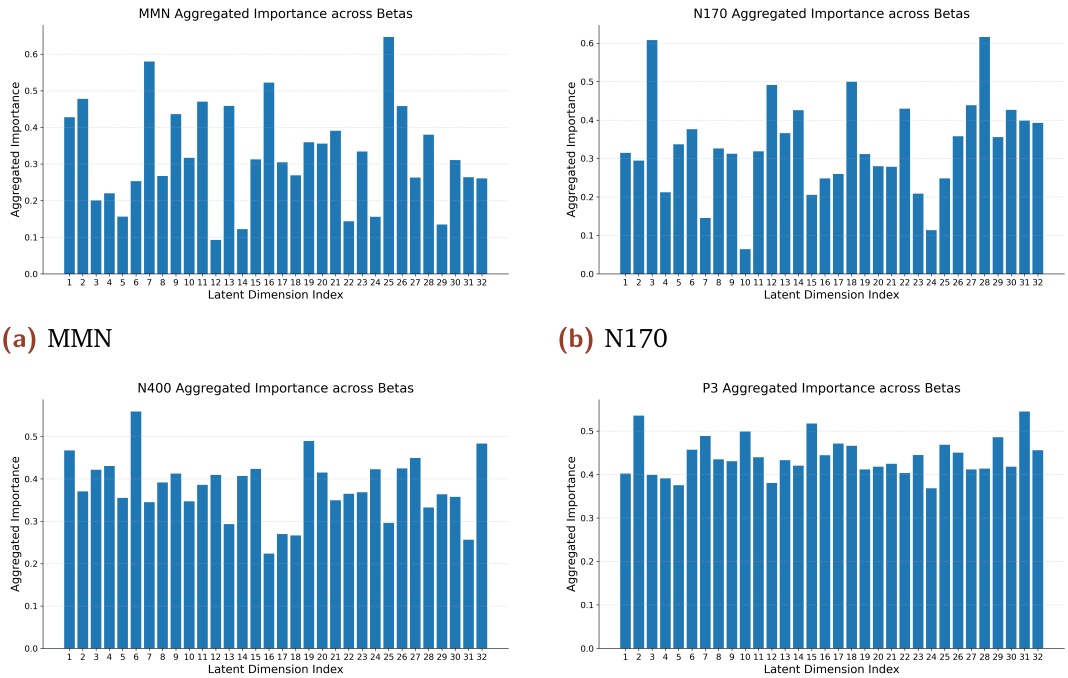
(d) Faces-scrambled

Figure 3.15.: Time-frame importance derived from time-frame masking for four β coefficients of N170, segment number $k = 100$.

is between 150-300ms, while under the 'cars-scrambled' condition, the peak appears at the beginning, and under the 'faces-scrambled' condition, the distribution is very uniform.

3.4 Ablation

3.4.1 Latent Ablation



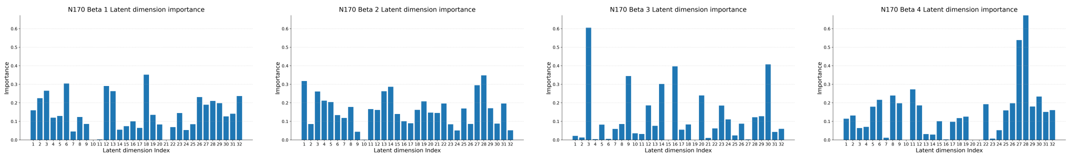
(a) MMN **(b)** N170

(c) N400 **(d)** P3

Figure 3.16.: Latent dimension importance derived from latent ablation for four ERP components. The importance score is normalized within each β range.

Figure 3.16 shows the importance of each latent dimension derived from the latent ablation analysis. The importance distribution of latent dimensions is not uniform across all ERP components, indicating that different latent units contribute differently to the reconstruction performance of different ERP components.

For MMN and N170, some latent dimensions are significantly more important than others, such as Latent 7 and Latent 25 in MMN, Latent 3 and Latent 28 in N170, while others are significantly less important, indicating a stronger dependence on specific latent features for these ERP components. In contrast, the importance values of N400 and P3 are more evenly distributed across the latent dimensions, with fewer peaks. This result indicates that their reconstruction relies on a broader combination of latent dimensions, rather than a few highly influential dimensions.



(a) Cars **(b)** Faces **(c)** Cars-scrambled **(d)** Faces-scrambled

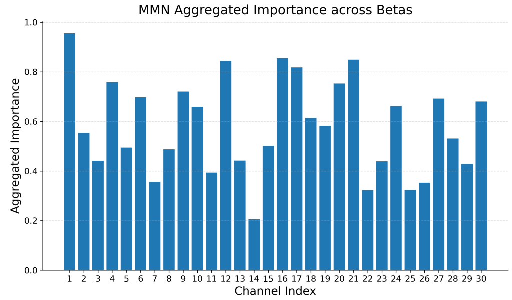
Figure 3.17.: Latent dimension importance for four β coefficients of N170.

Interestingly, the importance of latent dimensions varies significantly for different β coefficients within the same ERP component. In some cases, certain latent dimensions contribute little or nothing to the reconstruction under some specific β conditions. For example, considering the four β coefficients of N170 (Figure 3.17), latent dimensions 10 and 21 do not contribute to the reconstruction under the "car" condition. Moreover, some latent dimensions that are very important under the "cars" condition play only a minor role under the "car-scrambled" condition, indicating that different experimental conditions rely on different subsets of the latent space.

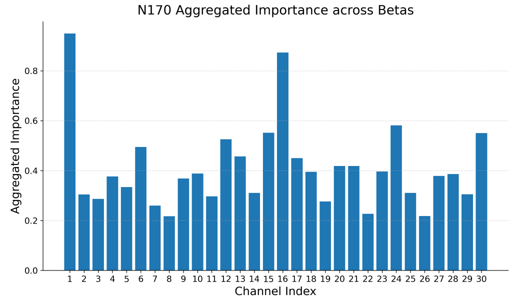
Overall, these results indicate that some dimensions of the learned latent representations play a more important role in some ERP components. Meanwhile, some dimensions may contribute nothing to the reconstruction, suggesting they haven't learned any features about that β coefficient. These also demonstrate that the ERP-related information is distributed across multiple latent dimensions, and no single latent dimension dominates in all components.

3.4.2 Channel Ablation

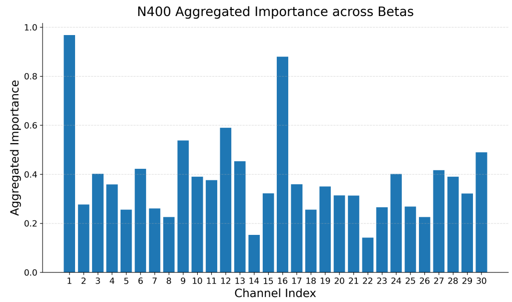
Figure 3.18 shows the channel importance results obtained from channel ablation analysis. We can observe a significant non-uniformity in the distribution of channel importance in the results of all ERP components, indicating that removing different channels and their corresponding connected edges has varying degrees of impact on the model reconstruction performance. This result shows that the model does not learn equal information from all channels, but rather relies more heavily on information from some key channels.



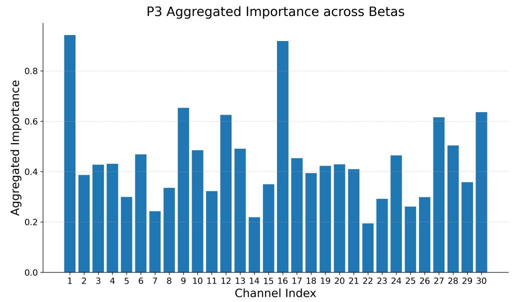
(a) MMN



(b) N170



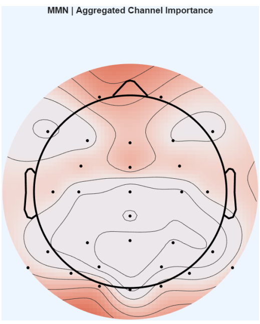
(c) N400



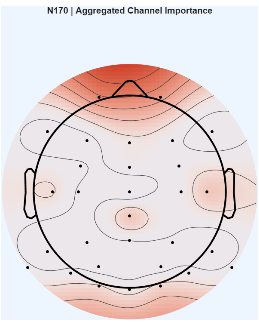
(d) P3

Figure 3.18.: Channel importance derived from channel ablation across β coefficients. The importance score is normalized within each β range.

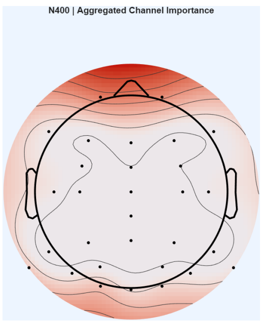
Among multiple ERP components, some channels consistently exhibit high importance, including 1 (FP1), 9 (P9), 12 (O1), 16 (FP2), and 30 (O2). Conversely, some channels consistently showed very low importance, such as 7 (P3), 14 (Pz), and 22 (Cz). This result is very similar to, but not entirely identical to, the previous channel masking results (Figure 3.6).



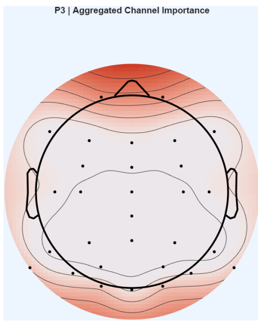
(a) MMN



(b) N170



(c) N400



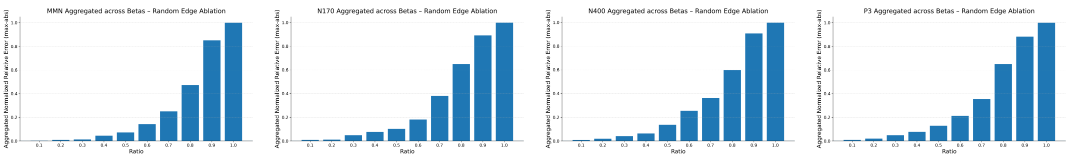
(d) P3

Figure 3.19.: Scalp topography of channel ablation results for the four ERP components.

Comparing the scalp distributions obtained from channel ablation (Figure ref3.19) and channel masking (Figure 3.9) further reveals their similarity, with their important channels both mainly located at the edge of the scalp. However, the distributions are not entirely the same. For MMN, channel masking has little effect on the frontal lobe region, while channel ablation highlights the importance of the frontal lobe region. Similarly, for N170, ablation reveals the importance of the temporal lobe region, while channel masking does not. These differences may be because channel ablation removes the nodes themselves and their associated graph structure, while channel masking only perturbs channels but preserves their connections.

This cross-method consistency demonstrates the robustness of our method. Meanwhile, the difference between channel masking and channel ablation also highlights the role of graph connectivity, suggesting that some channels function primarily through their structural location in the graph rather than through their individual signal content when forming the GAE latent space, such as channel 6 (C5) and 24 (C6).

3.4.3 Edge Ablation



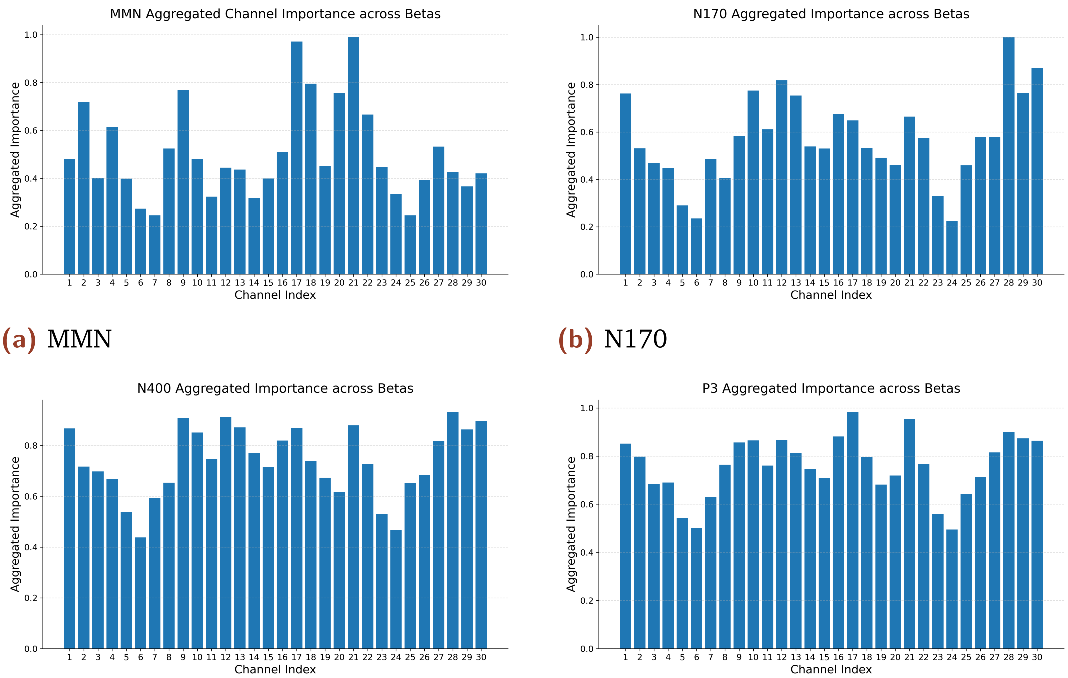
(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

Figure 3.20.: Random edge ablation results for four ERP components. The x-axis represents the proportion of edges removed from the graph, and the y-axis represents the normalized reconstruction error.

Figure 3.20 illustrates the impact of progressively random edge removal on the reconstruction performance of the GAE model. Across all ERP components, as the proportion of removed edges increases, the reconstruction error gradually increases. This consistent trend for all ERP components indicates that graph connectivity plays a crucial role in the model's ability to reconstruct ERP data.

At lower ablation ratios, the increase in reconstruction error is relatively small, suggesting that the model is robust to limited random edge removal. However, as the proportion of removed edges increases, reconstruction performance rapidly declines, indicating the importance of maintaining sufficient graph structure for information transmission.

Moreover, the similarity of the curves of the MMN, N170, N400, and P3 components suggests that the dependence on graph connectivity is a general characteristic of the model, rather than specific to any particular ERP component. This similarity further implies that when GAE learns these ERP components may similarly utilize the graph structure.



(a) MMN

(b) N170

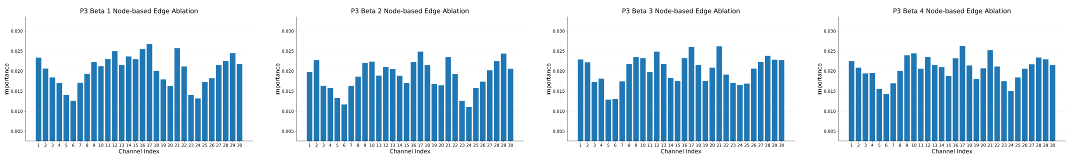
(c) N400

(d) P3

Figure 3.21.: The importance of channel-related edges derived from node-based edge ablation across β coefficients. The importance score is normalized within each β range, and then all results are summed and averaged.

Figure 3.21 illustrates the results of node-based edge ablation analysis. This analysis is independent of the signal characteristics of each channel and studies the contribution of each channel's connectivity individually. The importance of channels' connectivity varies significantly across all ERP components, indicating that removing edges associated with different nodes has varying impacts on reconstruction performance.

Some channels show high importance across all components, such as channels 17 (Fz) and 21 (FCz), suggesting that their connectivity plays a crucial role in information transmission within the graph (Figure 2.1). In contrast, other channels have relatively low importance, indicating their limited contribution through associated edges.

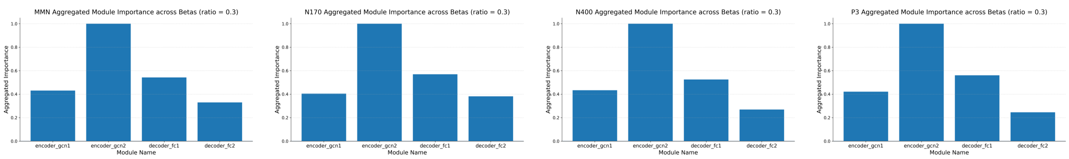


(a) AA target **(b)** AB non-target **(c)** AC non-target **(d)** AD non-target)

Figure 3.22.: The importance of channel-related edges derived from node-based edge ablation for the first four β coefficients of P3.

Interestingly, the importance distributions of all β coefficients for all ERP components are similar. For example, as shown in Figure 3.22, the first four β coefficients of P3 show a highly consistent importance distribution. Notably, these distribution patterns are very similar to those obtained using channel masking (Figure 3.6). This consistency suggests that GAE fully utilizes the structure of the graph, learns common spatial patterns in ERPs.

3.4.4 Module Ablation

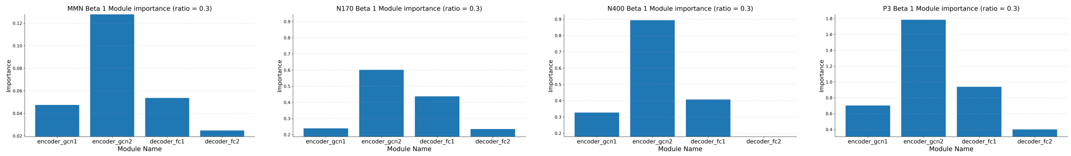


(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

Figure 3.23.: Module ablation results for four ERP components with ratio = 0.3. For each ERP component, the importance score is normalized within each β range, and then all results are summed and averaged.

Figure 3.23 shows the aggregated results of the module ablation of each ERP component module. Among all ERP components, ablation of different GAE modules leads to varying degrees of decrease in reconstruction performance, indicating that the contributions of each GAE module are not uniform.

In particular, the ablation of the second graph convolutional module ('$encoder\_gcn2$') in the GAE results in larger reconstruction errors compared to other modules. This means that this module plays a more significant role in learning information for all ERP components. In contrast, ablations of other modules have a relatively smaller influence on reconstruction performance, indicating that they play a supporting role in the overall architecture. This result is consistent with our findings in the previous Linear Probing section, which shows that the latent representation contains sufficient information and plays the most important role within the entire GAE model.



(a) MMN-Standard tone **(b)** N170-Faces **(c)** N400-prime,related,list1 **(d)** P3-AA target

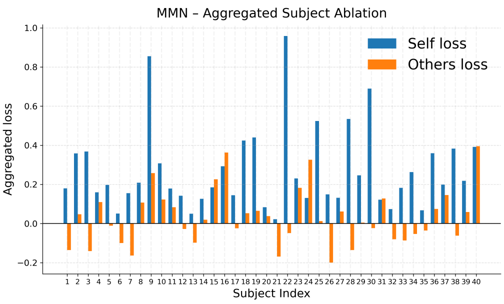
Figure 3.24.: Example module ablation results for the first beta parameter of four ERP components with ratio = 0.3. Other β coefficients exhibit highly similar module importance patterns and are therefore not shown.

Importantly, similar module importance patterns were observed under different beta coefficients, as shown in Figure 3.24. Similar patterns were also observed in other ERP components, with the module importance ranking as follows: $encoder\_gcn2 > decoder\_fc1 > encoder\_gcn1 > decoder\_fc2$. This indicates that the relative contribution of each GAE module is stable and not specific to any particular experimental condition. Module '$encoder\_gcn2$' is the most important because it contains the latent representations.

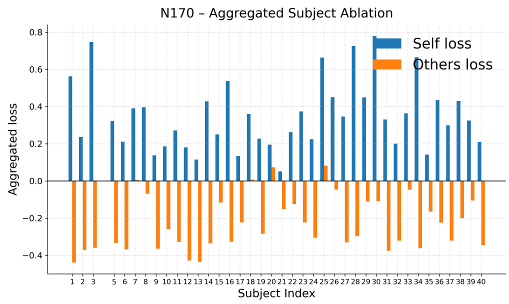
3.4.5 Subject Abaltion

From Figure 3.25 we can see that, across all ERP components, the impact of ablating a specific subject on reconstruction performance varied, indicating that subjects did not contribute equally to the learned representations. For the vast majority of subjects, ablation led to an increase in the reconstruction error of the ablated subject (self-loss), reflecting the model's dependence on subject-specific information.

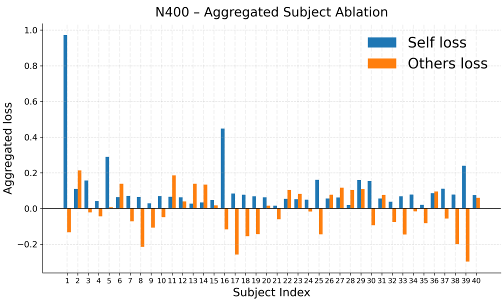
In contrast, the impact of ablating one subject on the reconstruction performance of other subjects (other subject loss) is typically small and often negative, which means that removing one subject from the data does not significantly reduce, and may even slightly improve, the reconstruction performance of the remaining subjects. This phenomenon suggests that the model primarily captures ERP information shared among subject data, rather than relying on unique information provided by individual subjects. Furthermore, the fact that removing one subject has only a small impact on other subjects demonstrates



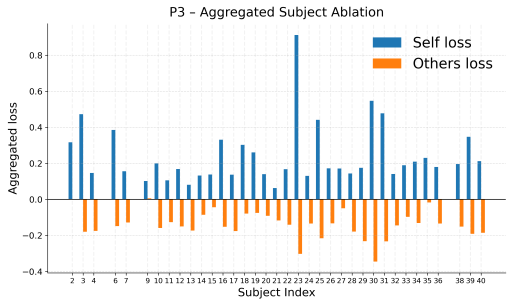
(a) MMN



(b) N170



(c) N400



(d) P3

Figure 3.25.: Subject importance derived from subject masking across ERP components. The importance score is normalized within each β range. The blue bars ("Self loss") indicate the change in reconstruction error for the masked subject itself, while the orange bars ("Others loss") represent the average change in reconstruction error for the remaining subjects. Note that due to data quality issues, Subject 4 was excluded for N170, and Subjects 1, 5, 8, and 37 were excluded for P3 in the GAE-based reconstruction. Therefore, the corresponding plots do not contain data for these subjects.

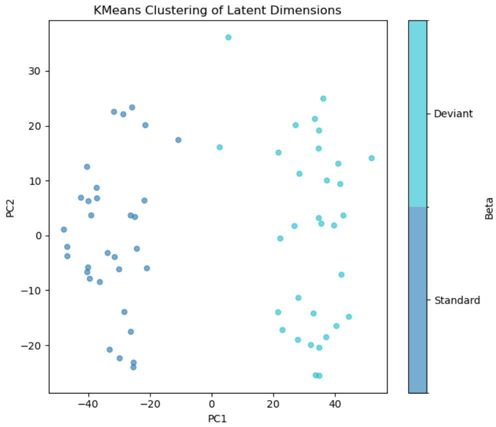
that the latent representations learned by GAE are not dominated by any single subject, showing the model's robustness to subject-level perturbations.

Moreover, a small number of subjects, such as Subject 22 in MMN and Subject 23 in P3, have extremely high self-loss values and a small impact on the reconstruction of other subjects' data. This suggests that these subjects may contain a lot of unique information, which leads to a stronger influence on the learning of latent representations. Without this information, the model can not reconstruct their data well. This inter-subject difference may reflect variations in data quality, signal-to-noise ratio, or individual participant characteristics.

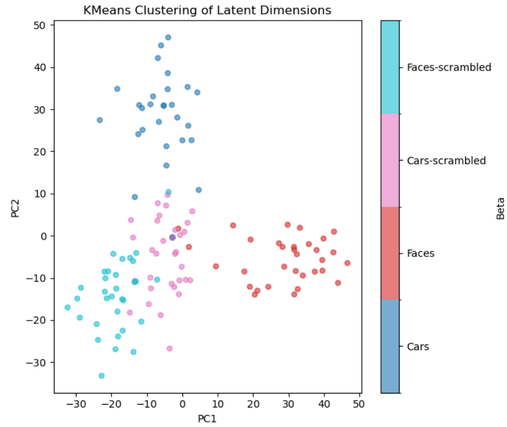
Meanwhile, comparing the subject weights used for 2nd-level analysis shown in Figure 3.3, we can see that the distributions of the results are inconsistent. Some subjects show great importance in the ablation results but have very small weights in the subject weight results. It can be divided into two cases: First, like Subject 22 in MMN, its weight in 2nd-level analysis is very small, and the ablation results show that its self-loss is extremely large, while its other loss is even reduced. This indicates that this subject has a strong influence on the model in its own data structure, but its representation is not shared with other subjects and may even have an interfering effect at the group level. Second, like Subject 40 in MMN, its self-loss and other loss are both large, but its weight in the 2nd-level analysis is small. This indicates that this subject does not represent a typical MMN pattern in the group statistics, but its data provides key information for GAE to learn latent representations.

3.5 Activation Maximization

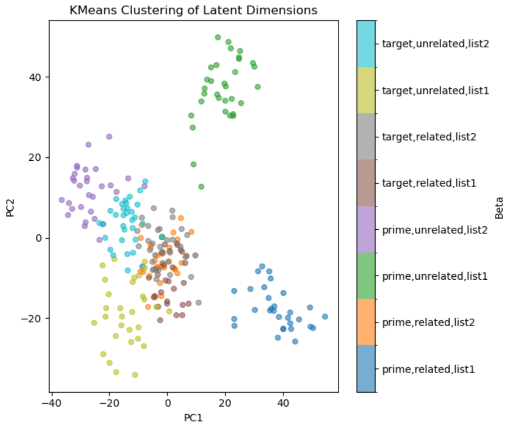
Figure 3.26 illustrates the distribution of latent dimensions based on activation maximization across different β. Within each ERP component, the latent dimensions from different β coefficient models exhibit both separation and overlap in the projection space. The separation between points indicates that models trained on different β parameters tend to learn distinct internal representations, indicating that β-specific information influences the structure of the learned latent features. For example, in the MMN case, the latent dimensions corresponding to the "Standard" and "Deviant" conditions have



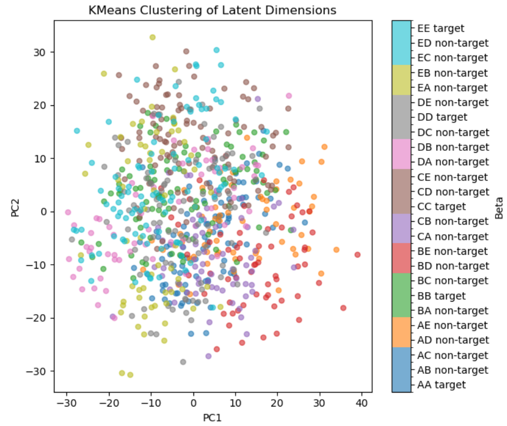
(a) MMN



(b) N170



(c) N400



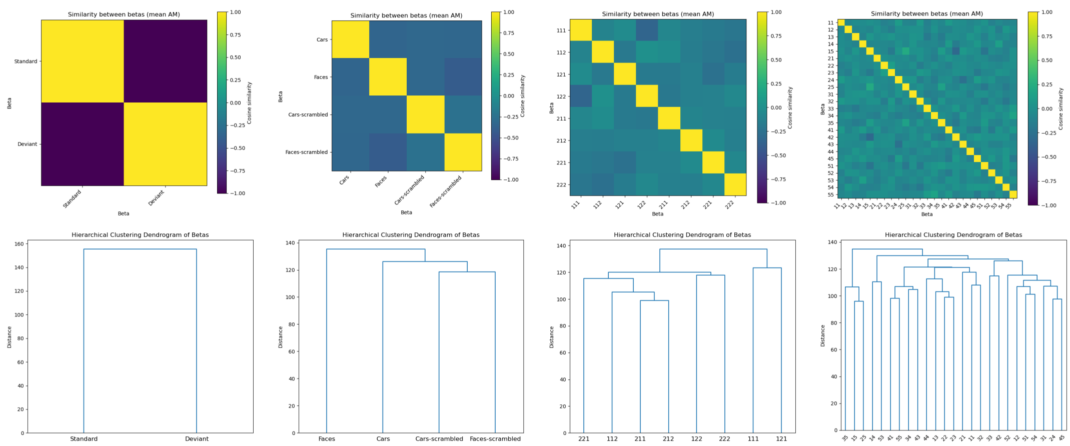
(d) P3

Figure 3.26.: Activation-maximization–based distribution of latent dimensions across β-specific GAE models. Each point represents a latent dimension of a β-specific model and is color-coded according to the β. Points that are closer to each other correspond to latent dimensions with more similar activation-maximizing input patterns, while points that are farther apart represent more different patterns. Detailed information about each β coefficient can be found in the Appendix A.

almost no overlap, indicating that the activation maximization (AM) patterns learned by their respective GAE models are significantly different.

At the same time, the observed overlap of nodes between different β coefficients suggests that certain activation maximization patterns are shared, reflecting a common spatiotemporal structure present in the ERP data. For example, in the N170 case, the latent dimensions corresponding to the "Cars-scramble" and "Faces-scrambled" conditions have a large overlap, indicating that the activation maximization (AM) modes learned by their respective GAE models have a common or similar spatiotemporal characteristic.

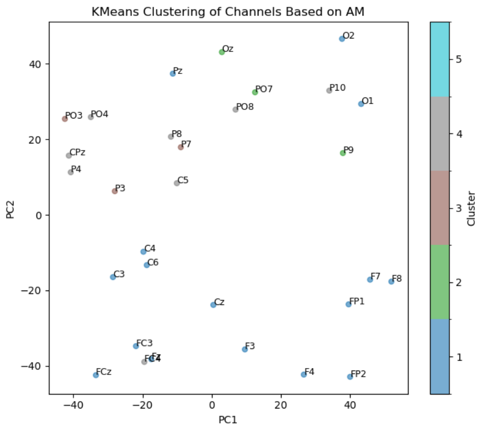
Moreover, clustering of points belonging to the same β indicates that latent dimensions within the model tend to produce activation maximization patterns with similar structures. This suggests that the model learns a set of latent features specific to the β coefficient, even though previous latent ablation studies found that different latent dimensions contribute differently; their "activation pattern style" is similar.



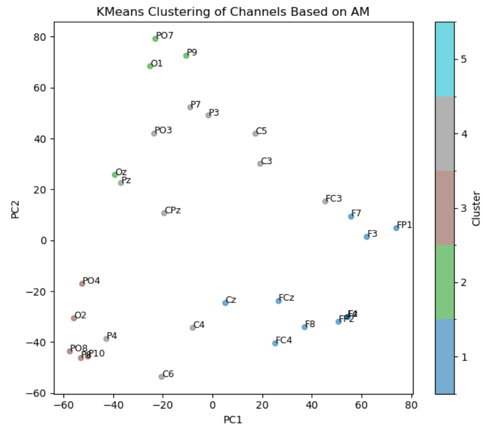
(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

Figure 3.27.: Similarity and hierarchical clustering based on mean activation-maximized patterns. The top figures show the cosine similarity matrix between betas, and the bottom figures show a hierarchical clustering dendrogram, illustrating the similarity relationship of the maximum activation patterns between specific beta models. Merging at lower beta levels indicates greater similarity, while merging at higher beta levels indicates less similarity.

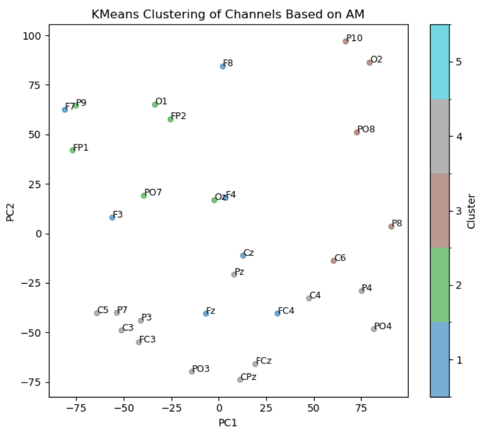
Figure 3.27 illustrates the similarity relationships between the average activation maximization (AM) patterns for each β coefficient in two additional forms. Among all ERP components, the cosine similarity matrix exhibits a clear structure rather than uniform similarity, indicating that the representations learned by the β-specific models differ significantly. Meanwhile, from the dendrogram, we can see that some pairs of β coefficients show relatively high similarity. The similarity between the activation maximization modes of different models indicates that these models rely on partially similar latent representation space structures for reconstruction. This indicates that although β-specific models are trained independently, they are able to capture common ERP-related features, reflecting common underlying features present under different experimental conditions.



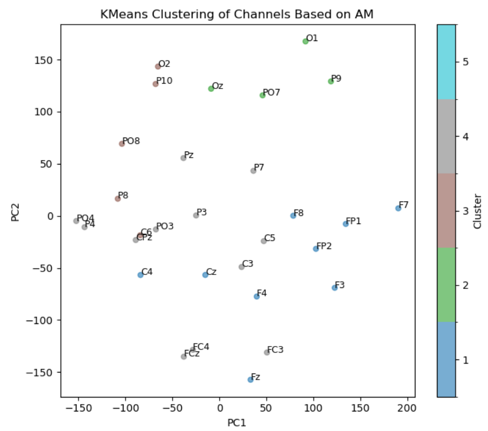
(a) MMN



(b) N170



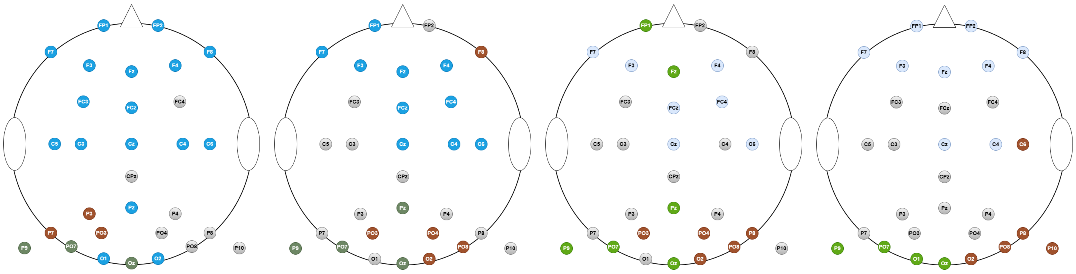
(c) N400



(d) P3

Figure 3.28.: Channel clustering based on activation-maximized patterns. The activation-maximized patterns are standardized and clustered using KMeans (k = 5), each point corresponding to one EEG channel and colors indicating cluster membership.

Figure 3.28 shows the clustering of EEG channels based on their activation-maximized patterns. And Figure 3.29 shows the clustering results on the scalp image. Regional clustering of channels can be observed on the scalp map,



(a) MMN **(b)** N170 **(c)** N400 **(d)** P3

Figure 3.29.: Scalp topography of channel-wise activation maximization (AM)–based clustering for the four ERP components.

indicating that adjacent channels tend to exhibit similar activation maximization (AM) patterns. This phenomenon indicates that GAE captures structured, region-specific latent representations, rather than treating channels as independent units. Importantly, these similarities are activation-maximizing similarities, thus reflecting the model's internal representational preferences rather than the similarity of the original ERP waveforms between channels.

4 Discussion

This thesis investigates a graph autoencoder (GAE)-based framework for weighting event-related potential (ERP) data in group-level statistical analysis and systematically studies what the GAE model learns using a series of explainable artificial intelligence (xAI) techniques.

The results of this study demonstrate that the GAE-based weighting method can achieve the expected results. The learned weights effectively modify the participants' contributions in group-level analyses, improve statistical robustness, and the weighting process does not produce fake effects or reduce real ERP effects. Meanwhile, 2nd-level analyses using the GAE weighted method can reproduce the classic ERP effects of MMN, N170, N400, and P3, including their typical spatiotemporal characteristics.

When using the XAI technique to compare the effects of different elements on the formation of latent space in GAE, we find that the model does not rely entirely on significant time windows or representative channels of ERP effects. Instead, some marginal or generally considered less important channels and time frames also play an important role. This shows that GAE makes full use of those channels that provide stable spatial structures, while relying on a wider range of temporal structures that support robust reconstruction.

Meanwhile, we also observed consistency across different methods and ERP components, with similar spatial importance distribution appearing in channel masking, channel ablation, and edge ablation. The important channels that the GAEs depend on remained stable under different β coefficients of the same ERP and showed significant overlap across different ERP components and analysis methods, indicating that GAEs rely on shared spatial structures within different β coefficients and ERP components. Activation maximization analysis further confirms this cross-method and cross-component consistency. We find that latent dimensions within the same β-specific model produce highly similar activation maximization patterns, while models trained based on different β coefficients also exhibit clustering. Meanwhile, we also observed overlap between β-specific clusters, suggesting that although different β-specific GAEs learn different latent representations, they capture information and structures shared across different ERPs components. These consistent findings suggest that GAEs do not depend on isolated elements or component-specific characteristics.

In conclusion, the results demonstrate that different components, such as subjects, channels, time frames, and model modules, contribute unequally to the formation of latent representations and the reconstruction of β coefficient data, and these differences reflect meaningful structure rather than noise. More importantly, the combination of masking, ablation, linear probing, and activation maximization analysis reveals consistent organizational patterns across multiple scales, demonstrating that GAE can aggressively learn and capture the stable spatiotemporal structure underlying ERP components.

However, several limitations need to be noted. First, the current GAE-based weighting method cannot be applied to N2pc, ERN, and LRP components because some subjects in these paradigms have different numbers of regressors, which means their GLM design matrices are not consistent. For example, in the ERN paradigm, some subjects make almost no mistakes, making the regressor for the "error" condition absent. This makes the GAE unable to learning from its data, as it can only accept data with the same shape. Second, training a separate GAE model for each β coefficient increases computational cost, so using xAI functionality would significantly increase runtime. Finally, although this study focuses on the well-defined ERP components in the ERP CORE, whether its results can be generalized to more complex EEG designs requires further investigation.

Future work could extend the proposed framework to joint modeling of multiple β coefficients or cross-paradigm representation learning, enabling the investigation of shared and paradigm-specific latent structures. Furthermore, the GAE model could be redesigned to handle components with inconsistent numbers of regressors, such as N2pc, ERN, and LRP, which is theoretically feasible. Finally, activation maximization (AM) could be explored more deeply. In this study, AM patterns were generated using a statistically regularized input space optimization method. Future work could employ generative activation maximization strategies, such as constrained optimization via deep generator networks (e.g., GANs or decoders), as proposed in previous work(Nguyen *et al.*, 2016), to generate AM prototypes with clearer structures.

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Appendix A

Index	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Name	FP1	F3	F7	FC3	C3	C5	P3	P7	P9	PO7	O1	Oz	Pz	CPz	FP2	Fz	F4	FC4	FCz	Cz	C4	C6	P4	P8	P10	PO8	PO4	O2	F8	PO3

Table A.1.: EEGLAB default channel indexing and corresponding electrode names for the 10–20 system used in this study.

The channel indexing shown in Table A.1 differs from the graph-based channel index used elsewhere in the thesis. Specifically, channels prior to index 10 retain identical ordering and indices across two channel indexings, while from index 10 onwards, the order remains unchanged, except for channels F8 and PO3, whose indices are reassigned (F8: 29→19; PO3: 30→11), which in turn changes the indices of the remaining channels.

Based on *ERP CORE: An open resource for human event-related potential research* (Kappenman *et al.*, 2021) and its Supplementary Materials, we summarize the four experimental paradigms-Passive Auditory Oddball Paradigm (MMN), Face Perception Paradigm (N170), Word Pair Judgement Paradigm (N400), and Active Visual Oddball Paradigm (P3) used in our study. Below, we provide a detailed description of them.

β index	Code	Trial type
1	80	Standard tone
2	70	Deviant tone

Table A.2.: MMN-Mapping between β indices and trial types.

Table A.2 demonstrates the correspondence between the β index and the trial type in the MMN task. Event codes represent tone intensity, measured in dB SPL. The standard tone intensity is 80 dB SPL, and the deviant tone intensity is 70 dB SPL. The standard tone occurs more frequently (p = 0.8), while the deviant tone occurs less frequently (p = 0.2), forming a passive auditory oddball paradigm designed to elicit MMN component.

Table A.3 demonstrates the correspondence between the β index and the trial type in the N170 task. Each β coefficient represents one of four visual stimulus categories: face, car, scrambled face, and scrambled car. Event

β index	Code	Trial type
1	faces	Faces
2	cars	Cars
3	faces-scrambled	Scrambled faces
4	cars-scrambled	Scrambled cars

Table A.3.: N170-Mapping between β indices and trial types.

types distinguish between object stimuli (faces and cars) and texture stimuli (scrambled faces and scrambled cars). In each trial, participants choose between two options by pressing buttons to determine whether the stimulus is an object or a texture.

β index	Code	Word position	Trial type	List
1	111	Prime	Related	List 1
2	112	Prime	Related	List 2
3	121	Prime	Unrelated	List 1
4	122	Prime	Unrelated	List 2
5	211	Target	Related	List 1
6	212	Target	Related	List 2
7	221	Target	Unrelated	List 1
8	222	Target	Unrelated	List 2

Table A.4.: N400-Mapping between β indices and trial types.

Table A.4 demonstrates the correspondence between the β index and the trial type in the N400 task. In the N400 task, each trial consisted of a prime word (in red) followed by a target (in green) word. Each β coefficient represents a specific event type, which is defined by the position of the words in the trial (prime vs. target word), the semantic relatedness between the prime and target word (related vs. unrelated), and the counterbalanced stimulus list (list 1 vs. list 2). The three-digit code means: the first digit indicates the position of the word in the trial (1 = prime word; 2 = target word), the second digit indicates the semantic relevance between the prime word and the target word (1 = related; 2 = unrelated), and the third digit indicates the list (1 = list1; 2 = list2). The target word is presented once in each of two counterbalanced stimulus lists, one in a semantically related context and the other in an unrelated context.

Table A.5 demonstrates the correspondence between the β index and the trial type in the P3 task. Each β is defined by the target letter of the current block and the letter presented in a given trial. The numbers 1-5 in the code correspond to the letters A-E respectively (1=A, 2=B, 3=C, 4=D, 5=E). In each block, one letter is designated as the target, and the other four letters are designated as non-targets. Target trials correspond to codes where the block

β index	Code	Block target	Trial stimulus	Trial type
1	11	A	A	Target
2	12	A	B	Non-target
3	13	A	C	Non-target
4	14	A	D	Non-target
5	15	A	E	Non-target
6	21	B	A	Non-target
7	22	B	B	Target
8	23	B	C	Non-target
9	24	B	D	Non-target
10	25	B	E	Non-target
11	31	C	A	Non-target
12	32	C	B	Non-target
13	33	C	C	Target
14	34	C	D	Non-target
15	35	C	E	Non-target
16	41	D	A	Non-target
17	42	D	B	Non-target
18	43	D	C	Non-target
19	44	D	D	Target
20	45	D	E	Non-target
21	51	E	A	Non-target
22	52	E	B	Non-target
23	53	E	C	Non-target
24	54	E	D	Non-target
25	55	E	E	Target

Table A.5.: P3-Mapping between β indices and trial types.

target and trial stimulus are the same (e.g., 11, 22, 33, 44, 55), and occur with a low probability (p = 0.2), while all other codes represent non-target trials (occurring more frequently, p = 0.8).